Biostatistics lectures

2ND BIOCHEMISTRY DEPARTMENT

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Table of Contents

Chapter 1: sampling distribution

➯ Introduction of sampling distribution

عند عمل تجربة أو دراسة معينه (كدراسة نسبة المدخنين في مصر)، من المستحيل غالبًا أن نقيس أفراد المجتمع (Population) بالكامل.
فبالتالي نأخذ عينة (Sample) صغيرة من المجتمع فهي مجموعه جزئية من المجتمع وتكون هي محل الدراسة بينما المجتمع ككل يكون هو هدف الدراسة .

ويوجد مقاييس إحصائية تصف خواص المجتمع ككل (Population Parameters)

ومقاييس تصف خواص العينة (Sample Statistics)

Population Parameters [N] مقاييس المجتمع

حيث [N] هي عدد عناصر المجتمع Population size

Mean μ المتوسط

[mu]

$$\mu = \frac{\sum X_i}{N}$$ (نفس وحدة المجتمع)

Variance σ^2 التباين

[سيجما تربيع]

$$\sigma^2 = \frac{\sum(X_i - \mu)^2}{N}$$ (مربع وحدة المجتمع)

Standard Deviation σ الانحراف المعياري

[سيجما]

$$\sigma = \sqrt{\sigma^2}$$ (نفس وحدة المجتمع)

Sample Statistics [n] مقاييس العينة

حيث [n] هي عدد عناصر العينة sample size

Mean $\overline{X}$ المتوسط

[X bar]

$$\overline{X} = \frac{\sum X_i}{n}$$ (نفس وحدة المجتمع)

Variance s^2 التباين

$$s^2 = \frac{\sum(X_i - \overline{X})^2}{n-1}$$ (مربع وحدة المجتمع)

Standard Deviation s الانحراف المعياري

$$s = \sqrt{s^2}$$ (نفس وحدة المجتمع)

حيث

المتوسط : المتوسط الحسابي لجميع القيم

التباين : متوسط مربعات الانحرافات لكل القيم عن متوسطها

الإنحراف المعياري : يعطينا فكرة عن متوسط بعد البيانات عن متوسطها. (مقدار التشتت)

التباين والانحراف المعياري نفس المفهوم (قياس مقدار التشتت) ولكن التباين بمربع وحدة القياس والانحراف المعياري بنفس وحدة القياس

X_i :مجموع عناصر المجتمع أو العينة

$X_i = X_1 + X_2 + X_3 .... + X_n$

مثال :

احسب متوسط الطول لدى المجتمع المصري أذا أخذت عينه بحجم ٧ وكانت الأطوال كالتالي :

1.65 , 1.70 , 1.75 , 1.85 , 1.60 , 1.8 , 1.9 m

Find the mean and variance and Standard Deviation

الحل :

$$\bar{X} = \frac{\sum X_i}{n} = \frac{1.65 + 1.70 + 1.75 + 1.85 + 1.60 + 1.8 + 1.9}{7} = 1.75\, m$$

$$s^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$$

$$s^2 = \frac{(1.65-1.75)^2 + (1.70-1.75)^2 + (1.75-1.75)^2 + (1.85-1.75)^2 + (1.60-1.75)^2 + (1.8-1.75)^2 + (1.9-1.75)^2}{6}$$

$$s^2 = 0.01167\, m^2$$

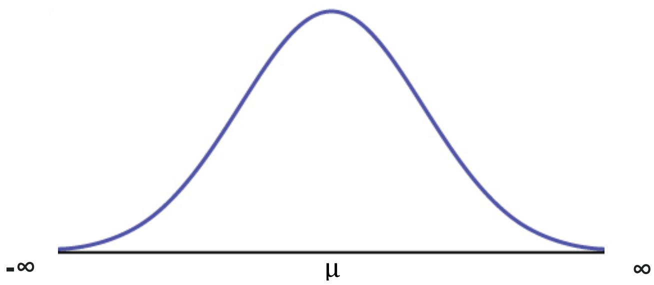
$$s = \sqrt{s^2} = \sqrt{0.01167} = 0.1\, m$$

Normal distribution
التوزيع الطبيعي

$f_{(x)}$ probability density function

داله كثافة الإحتمال

$$f_{(x)} = \frac{1}{\sqrt{2\pi\, \sigma^2}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad (-\infty < x < \infty)$$



منحنى التوزيع الطبيعي

خصائص منحنى التوزيع الطبيعي :

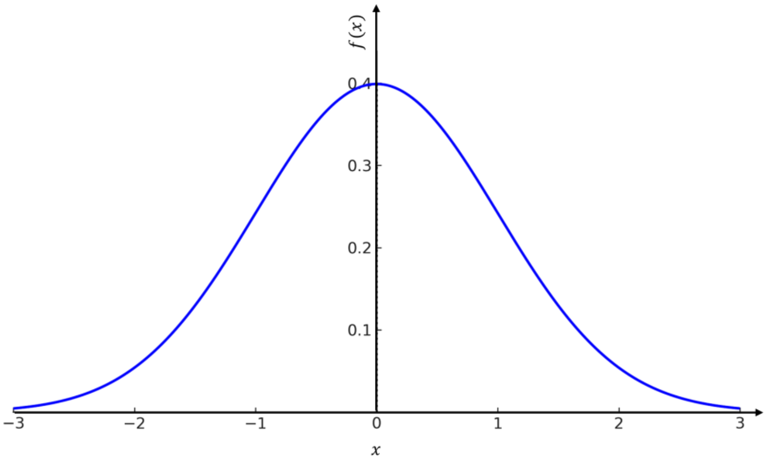
١. المنحنى ناقوصي الشكل (شبه الجرس)
٢. مداه من ∞ - الى ∞ [نهايتاه تتقاربان من الخط الأفقي أو من المحور ∞ , ∞- ولا يتلامسان معه]
٣. المساحة تحت المنحنى ثابته = الواحد الصحيح
٤. المساحة متماثله حول المتوسط
٥. المتوسط يحدد موضع المنحنى
٦. الإنحراف المعياري σ يحدد عرض المنحنى

Standard Normal distribution
التوزيع الطبيعي القياسي

هي حالة خاصة من التوزيع الطبيعي حيث
($\mu = 0 \quad \sigma = 1$)

$$f_{(x)} = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\, Z^2} \qquad\qquad Z = \frac{x - \mu}{\sigma}$$

Standard Normal Distribution ($\mu = 0, \sigma = 1$)



لحساب احتماليه معينه كاحتمال (P (x < 3 نحسب المساحة تحت المنحنى والتي تعبر عن تلك الاحتمالية فمثلاً :

$$P(X < A) = \int_{-\infty}^{A} \left(\frac{1}{\sqrt{2\pi\,\sigma^2}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \right)$$

ونظراً لصعوبة التكامل على غير المتخصصين نحول التوزيع الى توزيع طبيعي قياسي كالتالي :

$$Z = \frac{x - \mu}{\sigma}$$

إذًا

$$P(X < A) = P\left(\frac{X - \mu}{\sigma} < \frac{A - \mu}{\sigma}\right) = P\left(z < \frac{A - \mu}{\sigma}\right)$$

وفي حالة التوزيع الطبيعي القياسي يمكن ايجاد قيمه الاحتمال من الجدول الموجود في الكتاب بعنوان
Areas Under The Standard Normal Curve
والجدول مقسم خطوط طوليه وخطوط عرضية
الخطوط العرضية تعبر عن قيمة الرقم الصحيح وأول رقم بعد العلامة العشرية
اما الخطوط الطولية تعبر عن قيمة ثاني رقم بعد العلامة العشرية
فالرقم المراد هو ناتج تقاطع الخط الطولي والخط العرضي

والجدول يعبر عن قيم $Z < A$ فقط

مثلاً : إذا كانت (P (Z < 1.26
فقيمة الاحتمال تكون في الجدول كالتالي:

Z	0.06
1.2	0.8962

وبالتالي
$P (Z < 1.26) = 0.8962$

حـالات هـامـه

- في حالة كان الاحتمال أكبر من يحسب كالتالي :

$$P(X > A) = 1 - P(X < A)$$

حيث أن المساحة كلها عبارة عن الواحد الصحيح فنحذف منه الجزء الغير مراد قياسه

- في حالة وجود $P(A < X < B)$ تحسب كالتالي :

$$P(A < X < B) = P(X < B) - P(X < A)$$

EX.

If x ~ N (μ = 25 , σ = 4)

1. P(x < 32)
2. P(x > 25)
3. P(30 < x < 37)

The answer

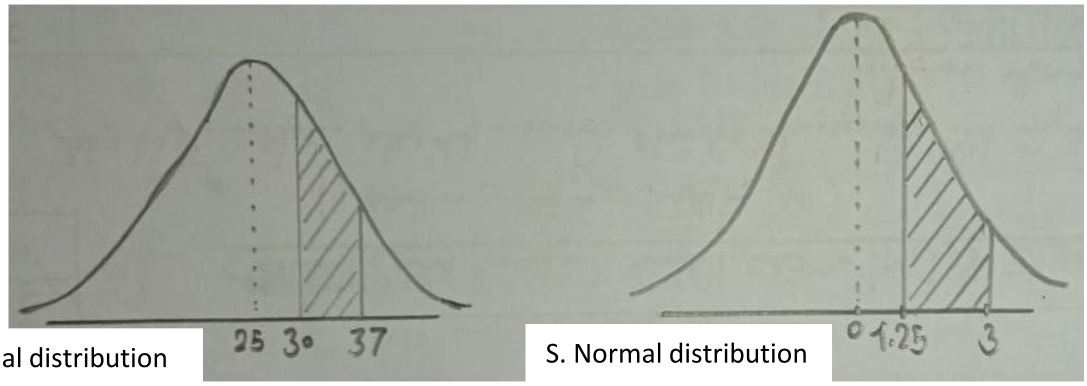
1. P(x < 32) = $P\left(\frac{X-\mu}{\sigma} < \frac{32-\mu}{\sigma}\right) = P\left(z < \frac{32-25}{4}\right) = P(z < 1.75)$ = 0.9599 [من الجدول]

2. P(x > 25) = 1 - P(x < 25) = 1 - $P\left(z < \frac{25-25}{4}\right)$ = 1 - $P(z < 0)$ = 1 - 0.5000 = 0.5

لاحظ هنا أنه بما أن المنحنى منحنى طبيعي قياسي فإن المساحتين حول المتوسط متساويتان وبما أن المساحة كلها = 1
وبما أن μ = 0 في المنحنى القياسي فإن إحتمال $P(z < 0)$ لابد أن يساوي 0.5 سواء أصغر أو أكبر من

3. P(30 < x < 37) = $P(X < 37) - P(X < 30) = P\left(Z < \frac{37-25}{4}\right) - P\left(Z < \frac{30-25}{4}\right)$

$$= P(Z < 3) - P(Z < 1.25) = 0.9987 - 0.8944 = 0.1043$$

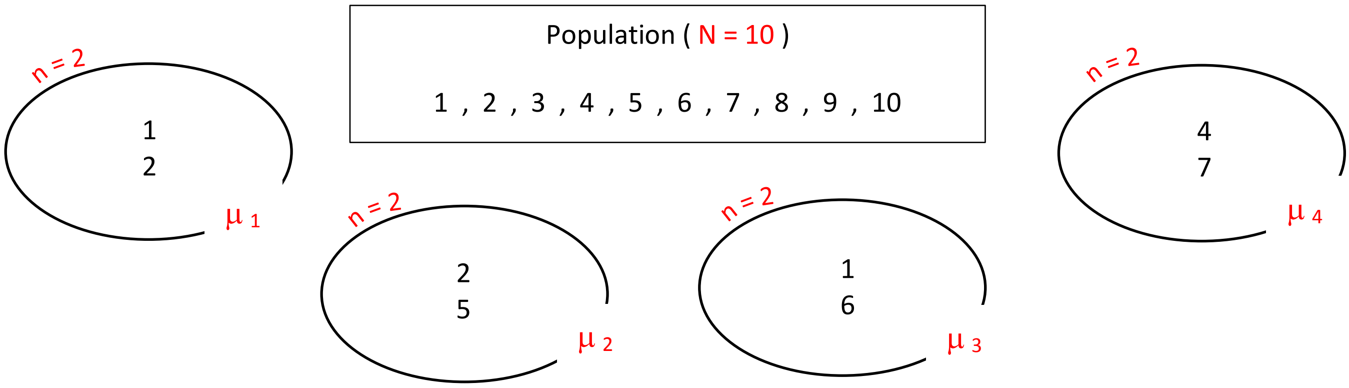


Normal distribution

S. Normal distribution

➪ sampling distribution of $\bar{X}$ and t – distribution

يمكن استخراج اكثر من عينه لمجتمع واحد فمثلا عند وجود مجتمع مكون من ١٠ ارقام وأردنا إستخراج عينه مكونة من عنصرين فيمكن في تلك الحالة استخراج أكثر من عينه كالتالي



ولكل عينة منهم متوسط لها يختلف عن العينه الأخرى

وبالتالي فإن متوسط تلك المتوسطات $\mu_{\bar{X}}$ يساوي مجموع جميع المتوسطات على عددها و هو يساوي متوسط المجتمع μ

$$\bar{X} \sim \bar{N}\left[\mu_{\bar{X}} = \mu \;,\; \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right]$$

و ذلك

في التوزيع الطبيعي

متوسط متوسطات العينات = متوسط المجتمع

والإنحراف المعياري لمتوسط العينات = الإنحراف المعياري للمجتمع مقسومًا على $\sqrt{n}$

وبالتالي

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\sigma \backslash \sqrt{n}}$$

الفرق بين القانونين هو $\sqrt{n}$

ولحساب عدد العينات الممكنه من مجتمع معين نستخدم القانون التالي :

$${}^{N}C_{n}$$

$${}^{10}C_{2} = \frac{10\,!}{2!\;8!} = 45$$

Sampling distribution of $\overline{X}$

the probability distribution of all possible sample means from a population

$$P(\overline{X} < a) = P\left(z < \frac{a-\mu}{\sigma \backslash \sqrt{n}}\right)$$

$$P(\overline{X} > a) = 1 - P\left(z < \frac{a-\mu}{\sigma \backslash \sqrt{n}}\right)$$

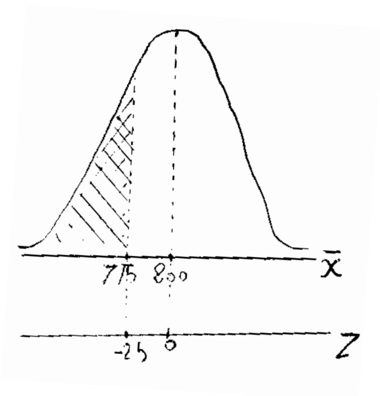
$$P(a < \overline{X} < b) = P(\overline{X} < b) - P(\overline{X} < a) = P\left(z < \frac{b-\mu}{\sigma \backslash \sqrt{n}}\right) - P\left(z < \frac{a-\mu}{\sigma \backslash \sqrt{n}}\right)$$

Example 6.1 [page 94]

An electric firm manufactures light bulbs that have a length of life that is approximately normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.

Solution:

$$\mu_{\overline{X}} = \mu = 800 \ , \ \sigma = 40 \ , \ n = 16$$



$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{16}} = 10$$

$$P(\overline{X} < 775) = P\left(z < \frac{775-\mu}{\sigma \backslash \sqrt{n}}\right) = P\left(z < \frac{775-800}{10}\right) = P(z < -2.5)$$

$$P(z < -2.5) = 0.0062$$

Z	0.00
- 2.5	0.0062

Additional questions

1. if $P(\overline{X} > 790)$

$$P(\overline{X} > 790) = 1 - P\left(z < \frac{790-\mu}{\sigma \backslash \sqrt{n}}\right) = 1 - P\left(z < \frac{790-800}{10}\right) = 1 - P(z < -1)$$

$$1 - P(z < -1) = 1 - 0.1587 = 0.8413$$

2. if $P(780 < \overline{X} < 810)$

$$P(780 < \overline{X} < 810) = P\left(z < \frac{810-\mu}{\sigma \backslash \sqrt{n}}\right) - P\left(z < \frac{780-\mu}{\sigma \backslash \sqrt{n}}\right) = P\left(z < \frac{810-800}{10}\right) - P\left(z < \frac{780-800}{10}\right)$$

$$= P(z < 1) - P(z < -2) = 0.8413 - 0.0228 = 0.885$$

Example :

The living spaces of all homes in a city have a normal distribution with mean 2300 square feet and standard deviation of 500 square feet

Let $\overline{X}$ mean living space for a random

Sample of 25 homes selected from this city , find

1. the probability $\overline{X}$ more than 2100
2. the probability $\overline{X}$ less than 2500
3. the probability $\overline{X}$ between 2700 , 2200

Solution:

$$\mu_{\overline{X}} = \mu = 2300 \ , \ \sigma = 500 \ , \ n = 25$$

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}} = \frac{500}{\sqrt{25}} = 100$$

1. $P(\overline{X} < 2100)$

$$P(\overline{X} > 2100) = 1 - P\left(z < \frac{2100 - 2300}{100}\right) = 1 - P(z < -2) = 1 - 0.0228 = 0.9772$$

2. $P(\overline{X} > 2500)$

$$P(\overline{X} < 2500) = P\left(z < \frac{2500 - 2300}{100}\right) = P(z < 2) = 0.9772$$

3. $P(780 < \overline{X} < 810)$

$$P(2200 < \overline{X} < 2700) = P\left(z < \frac{2700 - 2300}{100}\right) - P\left(z < \frac{2200 - 2300}{100}\right)$$

$$= P(z < 4) - P(z < -1) = 1 - 0.1587 = 0.8413$$

------------------------------------ ملاحظة هامة ------------------------------------

جدول الـ Z ينتهي بالقيمة 3.49 و 3.49 - حيث

$$P(Z < 3.49) = 0.9998 \cong 1$$
$$P(Z < -3.49) = 0.0002 \cong 0$$

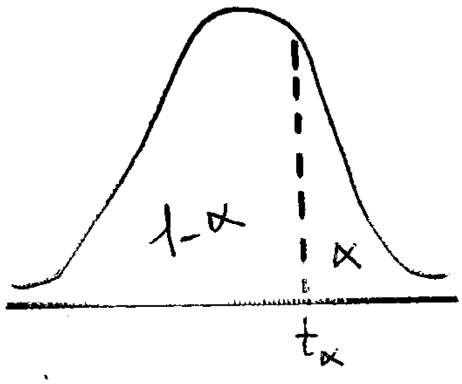
وبالتالي أي قيمة أكبر من 3.49 تساوي 1

وأي قيمة أصغر من 3.49- تساوي 0

Ex . $P(z < 4) = 1$

----

t – distribution



T_α ← تعني أن المساحة يمين تلك القيمة تساوي α وعلى يسارها 1 - α

$T_\alpha = -T_{1-\alpha}$ $\qquad\qquad$ $-T_\alpha = T_{1-\alpha}$

لتوزيع الـ T جدول خاص

ولمعرفة قيمة $T_{\alpha,\, df}$ يجب معرفة كلا من

قيمة α وهي المساحة يمين قيمة T

ودرجات الحرية Degree freedom [df]

df = n – 1

Example 6.2 [page 95]

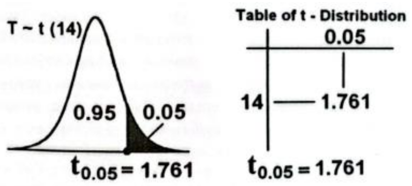
Find the t-value with v = 14 [df] that leaves an area of :

(a) 0.95 to the left.

(b) 0.95 to the right.

Solution:

(a) $T_{0.05} = 1.761$



(b) $T_{0.95} = -T_{0.05} = -1.761$

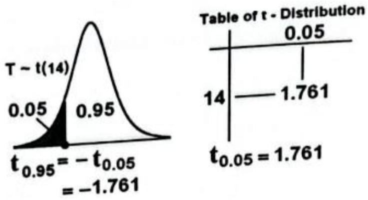
قيمة $\alpha = 0.95$ لا توجد في الجدول وبالتالي

نستخدم القانون السابق

$T_\alpha = -T_{1-\alpha}$

وبالتالي

$T_{0.95} = -T_{0.05}$



Chapter 2 : Estimation

⇨ Estimation of μ

عند وجود قيمة مجهولة من المجتمع نحتاج لتقدير قيمتها عن طريق عينة من هذا المجتمع
ويوجد طريقتان للتقدير وهما Point Estimation و Interval Estimation

Estimation of μ

Point Estimation

$$\hat{\mu} = \bar{X}$$

Interval estimation

(1 - α) 100 % C.I for μ

تعتمد على σ
حيث

Known

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

margin of error

Un Known

ننظر في تلك الحالة إلى حجم العينه

n ≥ 30

$$\bar{X} \pm Z_{\alpha/2} \frac{S}{\sqrt{n}}$$

n < 30

$$\bar{X} \pm t_{\alpha/2,\ df} \frac{S}{\sqrt{n}}$$

ملاحظات هامة

$\hat{\mu}$ (Mu hat) تعبر عن تقدير متوسط المجتمع

Interval estimation = Point Estimation ± margin of error

Interval estimation هي تقدير القيمة كفتره بين قيمتين بحيث نكون متيقنين أنه وبنسبة معينه ستكون قيمة المتوسط في تلك الفترة ونسبة تيقننا من تلك الفتره تسمى (مستوى الثقة level of confidance) وهي $(1 - \alpha) \times 100\ \%$

α ← level of significance

α تعبر عن المساحة يمين المنحنى وبالتالي لحساب $Z_{\alpha/2}$ فلإيجادها من الجدول الموجود في الكتاب والمعبر عن قيم إحتمال Z أصغر من فقط أي المساحة يسار قيمة Z نقوم بإيجاد قيمة $1 - \frac{\alpha}{2}$

وهذا سيأتي بيانه في الأمثلة

Example 6.3 [page 98]

The average zinc concentration recorded from a sample of zinc measurements in 36 different locations is found to be 2.6 gram/milliliter. Find a 95% and 99% confidence interval (C.I.) for the mean zinc concentration in the river. Assume that the population standard deviation is 0.3.

Solution :

$$n = 36 \ , \ \bar{X} = 2.6 \ , \ \sigma = 0.3$$

a. 95 % C.I. for μ

Since σ is known then
95 % C.I. for μ is

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{2}}$$

$$(1 - \alpha)100\% = 95\%$$

$1 - \alpha = 0.95$ $\qquad\qquad$ $\alpha = 0.05$

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$$

تلك هي المساحة على يمين قيمة Z المراد معرفتها
ولتحديد قيمة Z من الجدول نحتاج معرفة المساحة يسار القيمة لان الجدول يعبر عن إحتمال Z أصغر من فقط
ولمعرفة المساحة يسار القيمة نستخدم العلاقة

$$1 - \frac{\alpha}{2}$$

والمره دي هنجيب العكس من الإحتمال هنجيب قيمة Z

$1 - \frac{\alpha}{2} = 0.975$ $\qquad\qquad$ $Z_{\alpha/2} = Z_{0.025} = 1.96$

Z	0.06
1.9	0.975

$$95\ \%\ \text{C.I. for } \mu = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{2}} = 2.6 \pm 1.96 \frac{0.3}{\sqrt{36}}$$

[2.502 , 2.698]

a. 95 % C.I. for μ

$1 - \alpha = 0.99$ $\qquad\qquad$ $\alpha = 0.01$

$\frac{\alpha}{2} = \frac{0.01}{2} = 0.005$ $\qquad\qquad$ $1 - \frac{\alpha}{2} = 0.995$

Z	0.07	0.08
2.5	0.9949	0.9951

في حاله عدم وجود فيمه الإحتمال في الجدول ويوجد قيمتين وأصغر من القيمة المراده وكلاهما على نفس البعد منها نقوم بحساب قيمة Z من متوسط القيمتين

$$Z_{\alpha/2} = Z_{0.005} = \frac{2.57+2.58}{2} = 2.575$$

$$95\ \%\ \text{C.I. for } \mu = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{2}} = 2.6 \pm 2.575 \frac{0.3}{\sqrt{36}}$$

[2.471 , 2.729]

Example 6.4 [page 99]

A publishing company has just published a new college textbook. Before the company decides the price at which to sell this textbook, it wants to know the average price of all such textbooks in the market. The research department at the company took a sample of 36 such textbooks and collected information on their prices. This information produced a mean price of 48.40$ and the standard deviation of the prices textbooks is 4.50 $ for this sample.

(a) What is the point estimate of the mean price of all such college textbooks?
(b) Construct a 90% confidence interval for the mean price of all such college textbooks.

Solution:

$$n = 36\ ,\ \bar{X} = 48.40\ ,\ \sigma = 4.50$$

Since σ is known and n = 36 ≥ 30 , then
(1 - α) 100 % C.I for μ is

$$\bar{X} \pm Z_{\propto/2}\ \frac{s}{\sqrt{n}}$$

a. 95 % C.I. for μ

$$1 - \alpha = 0.90 \qquad \alpha = 0.1$$

$$\frac{\alpha}{2} = \frac{0.1}{2} = 0.05 \qquad 1 - \frac{\alpha}{2} = 0.95$$

$$Z_{0.1} = 1.645$$

Z	0.04	0.05
1.6	0.949	0.951

$$48.40 \pm 1.645\ \frac{4.5}{\sqrt{36}}$$

[47 . 17 , 49 . 63]

Example 6.6 [page 101]

The contents of 7 similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% C.I. for the mean of all such containers, assuming an approximate normal distribution.

Solution:

$$n = 7\ ,\ X = 9.8\,, 10.2\,, 10.4\,, 9.8\,, 10.0\,, 10.2\,, 9.6$$

$$\bar{X} = \frac{\sum x_i}{n} = \frac{9.8 + 10.2 + 10.4 + 9.8 + 10.0 + 10.2 + 9.6}{7} = 10$$

$$S = \sqrt{\frac{\sum (x_i - \bar{X})^2}{n-1}} = \sqrt{\frac{(9.8-10)^2 + (10.2-10)^2 + (10.4-10)^2 + (9.8-10)^2 + (10.0-10)^2 + (10.2-10)^2 + (9.6-10)^2}{6}} = 0.243$$

Since σ is known and n = 7 < 30 , then
95 % C.I for μ is

$$\bar{X} \pm t_{\alpha/2\,,\ df}\ \frac{s}{\sqrt{n}}$$

$1 - \alpha = 0.95 \qquad \alpha = 0.05$

$$\frac{\alpha}{2} = 0.025$$

v	α
	0.025
6	2.447

$df = n - 1 = 6$

$t_{\alpha/2\ ,\ df} = t_{0.025\ ,\ 6} = 2.447$

$$\overline{X} \pm t_{\alpha/2\ ,\ df} \frac{s}{\sqrt{n}}$$

$$10 \pm 2.447 \times \frac{0.243}{\sqrt{7}}$$

[9.74 , 10.26]

$P(9.74 < \mu < 10.25) = 95\% = 0.95$

➪ Estimating the difference between two means

عند وجود مجتمعين وأردنا تقدير الفرق بين متوسطات المجتمعين
وبالمثال يتضح المقال
إذا أردنا مثلاً إيجاد نسبة متوسط الطول في السودان و في مصر
وإيجاد النسبة بين متوسط الطول في السودان ومتوسط الطول في مصر

μ_1 السودان — n_1: $\bar{X}_1$, S_1

μ_2 مصر — n_2: $\bar{X}_2$, S_2

نقدر تلك النسبه بإستخدام عينة لكل من المجتمعين نستخدم القوانين التاليه
ويوجد طريقتان للتقدير وهما Point Estimation و Interval Estimation

Estimation of μ_1 - μ_2

Point Estimation

$$\hat{\mu}_1 - \hat{\mu}_2 = \bar{X}_1 - \bar{X}_2$$

Interval estimation

(1 - α) 100 % C.I for μ_1 - μ_2

تعتمد على σ_1 , σ_2

Known

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2}\sqrt{\frac{\sigma_1^{\,2}}{n_1} + \frac{\sigma_2^{\,2}}{n_2}}$$

Un Known

ننظر في تلك الحالة إلى n_1 , n_2

n1 , n2 $\geq$ 30

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2}\sqrt{\frac{S_1^{\,2}}{n_1} + \frac{S_2^{\,2}}{n_2}}$$

n1 , n2 < 30

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2,\ df}\ .Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

التباين المشترك $$Sp^2 = \frac{(n_1 - 1)\,S_1^{\,2} + (n_2 - 1)\,S_2^{\,2}}{(n_1 + n_2 - 2)}$$

$$V = df = n_1 + n_2 - 2$$

Example 6.7 [page 103]

An experiment was conducted in which two types of engines, A and B, were compared. Gas mileage in miles per gallon was measured. 50 experiments were conducted using engine type A and 75 experiments were done for engine type B. The gasoline used and other conditions were held constant. The average gas mileage for engine A was 36 miles per gallon and the average for engine B was 42 miles per gallon. Find 96% confidence interval for $\mu_B - \mu_A$, where μ_A and μ_B are population mean gas mileage for engines A and B, respectively. Assume that the population standard deviations are 6 and 8 for engines A and B, respectively.

Solution:

A

$n_1 = 50 \quad \overline{X}_1 = 75 \quad \alpha_1 = 6$

B

$n_2 = 75 \quad \overline{X}_2 = 42 \quad \alpha_2 = 8$

96% confidence interval for $\mu_B - \mu_A$

$1 - \alpha = 0.96 \qquad \alpha = 0.04$

$\frac{\alpha}{2} = 0.02 \qquad 1 - \frac{\alpha}{2} = 0.980$

Z	0.05	0.06
2.0	0.9798	0.983

نأخذ القيمة الأقرب للقيمه المرادة
وفي حالة تساوي بعد القيمتين عن القيمة المراده نأخذ المتوسط بينهما

$Z_{0.02} = 2.05$

$$(\overline{X}_2 - \overline{X}_1) \pm Z_{\alpha/2}\sqrt{\frac{\sigma_2{}^2}{n_2} + \frac{\sigma_1{}^2}{n_1}}$$

$$(42 - 36) \pm 2.05\sqrt{\frac{8^2}{75} + \frac{6^2}{50}}$$

[3.43 , 8.57]

Example 6.8 [page 104]

To compare the resistance of wire A with that of wire B, an experiment shows the following results based on two independent samples (original data multiplied by 1000):
Wire A: 140, 138, 143, 142, 144, 137
Wire B: 135, 140, 136, 142, 138, 140
Assuming equal variances, find 95% confidence interval for $\mu_A - \mu_B$, where μ_A and μ_B are the mean resistance of wire A and B.

Solution:

A

$$n_1 = 6 \quad X = 140, 138, 143, 142, 144, 137$$

$$\bar{X}_1 = \frac{\sum x_i}{n} = \frac{140 + 138 + 143 + 142 + 144 + 137}{6} = 140.67$$

$$S_1{}^2 = \frac{\sum (x_i - \bar{X}_1)^2}{n_1 - 1} = 7.8667$$

B

$$n_1 = 6 \quad X = 135, 140, 136, 142, 138, 140$$

$$\bar{X}_1 = \frac{\sum x_i}{n} = \frac{135 + 140 + 136 + 142 + 138 + 140}{6} = 138.5$$

$$S_2{}^2 = \frac{\sum (x_i - \bar{X}_2)^2}{n_2 - 1} = 7.1$$

$$Sp^2 = \frac{(n_1 - 1)\, S_1{}^2 + (n_2 - 1)\, S_2{}^2}{(n_1 + n_2 - 2)} = \frac{(6-1)\, 7.8667 + (6-1)\, 7.1}{10} = 7.48335$$

95% confidence interval for $\mu_A - \mu_B$

$1 - \alpha = 0.95 \qquad \alpha = 0.05$

$$\frac{\alpha}{2} = 0.025 \qquad df = n_1 + n_2 - 2 = 10$$

v	α
	0.025
10	2.228

$t_{0.025\,,\,10} = 2.228$

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2,\, df} \sqrt{\frac{Sp^2}{n_1} + \frac{Sp^2}{n_2}}$$

$$(140.67 - 138.5) \pm 2.228 \sqrt{\frac{7.48335}{6} + \frac{7.48335}{6}}$$

[- 1.35 , 5.69]

Example 6.10 [page 101]

Ann Howard conducted a management progress study with AT&T management job holders to assess the performance of those who possessed a college degree and those who did not. Let us refer to the two groups of participants as college and non-college participants, respectively. The samples included 274 college and 148 non-college participants. In the area of motivation for advancement, the mean scores were 2.89 for college participants and 2.70 for non-college participants, with the two standard deviations being 0.57 and 0.64, respectively. Find a 99% confidence interval for difference between the mean scores of the two respective populations in the area of motivation for advancement.

Solution:

College

$n_1 = 274 \quad \overline{X}_1 = 2.89 \quad S_1 = 0.57$

non-college

$n_2 = 148 \quad \overline{X}_2 = 2.70 \quad S_2 = 0.64$

99% confidence interval for $\mu_{college} - \mu_{non\text{-}college}$

$1 - \alpha = 0.99 \qquad \alpha = 0.01$

$\frac{\alpha}{2} = 0.005 \qquad 1 - \frac{\alpha}{2} = 0.995$

Z	0.07	0.08
2.5	0.9949	0.9951

$Z_{0.005} = 2.575$

$$(\overline{X}_1 - \overline{X}_2) \pm Z_{\alpha/2}\sqrt{\frac{{S_1}^2}{n_1} + \frac{{S_2}^2}{n_2}}$$

$$(2.89 - 2.70) \pm 2.575\sqrt{\frac{0.57^2}{274} + \frac{0.64^2}{148}}$$

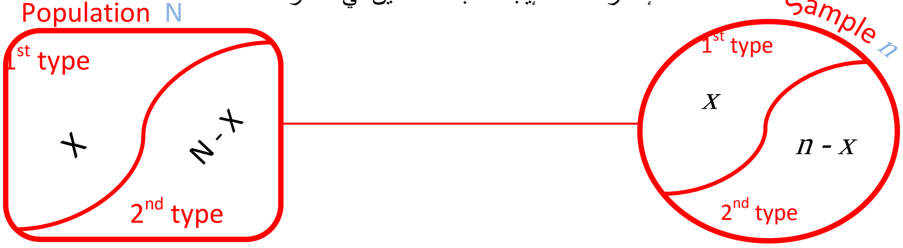
[0.0281 , 0.352]

⇨ Estimation of the population proportion

عند وجود نوعين من العناصر في المجتمع وأردنا حساب نسبة أحد العناصر الموجودة في المجتمع

وبالمثال يتضح المقال

إذا أردنا مثلاً إيجاد نسبة المدخنين في مصر



نقدر تلك النسبه بإستخدام العينة المأخوذة من المجتمع و نستخدم القوانين التاليه

ويوجد طريقتان للتقدير وهما Point Estimation و Interval Estimation

Estimation of p

Point Estimation

$$\hat{P} = \frac{x}{n}$$

Interval estimation

(1 - α) 100 % C.I for p

نسبة النوع المستهدف $$p = \hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}\ \hat{q}}{n}}$$

نسبة النوع الغير مستهدف $$\hat{q} = 1 - \hat{p}$$

Example 6.12 [page 108]

In a random sample of n = 500 families owing television sets in the city of Hamilton, Canada, it was found that x = 340 subscribed to HBO. Find 95% confidence interval for the actual proportion of families in this city who subscribe to HBO.

Solution:

$$n = 500 \quad x = 340$$

$$\hat{p} = \frac{x}{n} = \frac{340}{500} = 0.68$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.68 = 0.32$$

95% confidence interval for p

$$1 - \alpha = 0.95 \qquad \alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025 \qquad 1 - \frac{\alpha}{2} = 0.975$$

Z	0.06
1.9	0.9750

$$Z_{0.025} = 1.96$$

$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}\ \hat{q}}{n}} = 0.68 \pm 1.96 \times \sqrt{\frac{0.68 \times 0.32}{500}}$$

[0.64 , 0.72]

Example 6.14 [page 109]

A food company is planning to market a new kind of cereal. However, before marketing this product, the company wants to find what percentage of people will like it. The company's research department selected a random sample of 500 persons and asked them to taste this cereal. Of these 500 persons, 290 said they liked it.

(a) What is the point estimate of the population proportion?
What is the margin of error of this estimate?

(b) Find, with a 99% confidence level, what percentage of all people will like this cereal.

Solution:

$$n = 500 \quad x = 290$$

a. Point estimator $\hat{p} = \frac{x}{n} = \frac{290}{500} = 0.58$

$$\hat{q} = 1 - \hat{p} = 1 - 0.58 = 0.42$$

b. 99% confidence interval for p

$$1 - \alpha = 0.99 \qquad \alpha = 0.01$$

$$\frac{\alpha}{2} = 0.005 \qquad 1 - \frac{\alpha}{2} = 0.995$$

Z	0.07	0.08
2.5	0.9949	0.9951

$$Z_{0.025} = 2.575$$

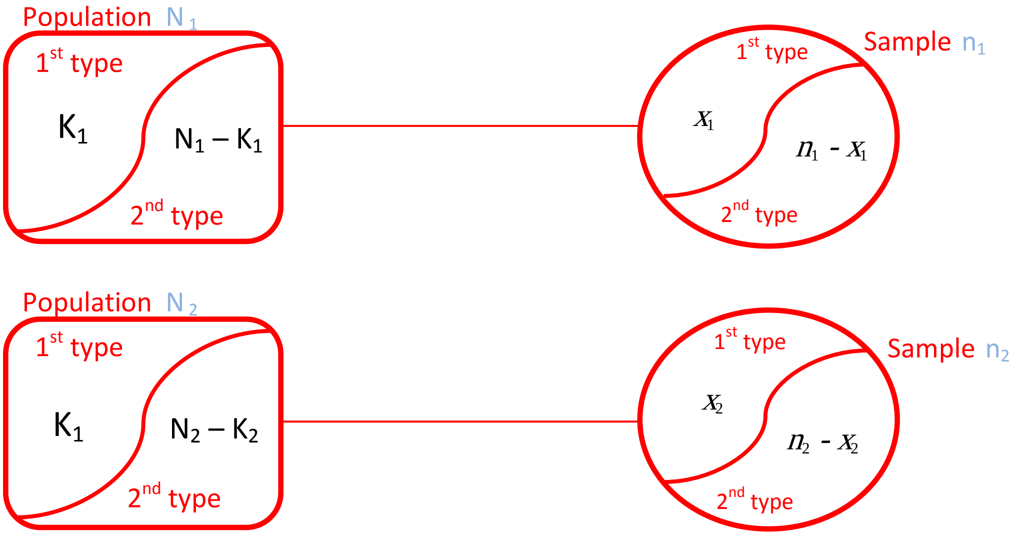
$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}\ \hat{q}}{n}} = 0.58 \pm 2.575 \times \sqrt{\frac{0.58 \times 0.42}{500}}$$

[0.52 , 0.64]

➪ Estimation the difference between two proportion

--- Two sample ---

إذا أردنا إيجاد الفرق بين نسبة المدخنين في مصر ونسبتهم في السودان نستخدم القوانين التاليه



Estimation of $p_{1} - p_{2}$

Point Estimation

$$\widehat{P}_1 - \widehat{P}_2$$

Interval estimation

(1 - α) 100 % C.I for p

$$\left(\widehat{P}_1 - \widehat{P}_2\right) \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \, \hat{q}_1}{n_1} + \frac{\hat{p}_2 \, \hat{q}_2}{n_2}}$$

$$\hat{q}_1 = 1 - \hat{p}_1$$

$$\hat{q}_2 = 1 - \hat{p}_2$$

Example 6.17 [page 112]

A certain change in a process for manufacture of component parts is being considered. Samples are taken using both existing and the new procedure to determine if the new process results in an improvement. If 75 of 1500 items from the existing procedure were found to be defective and 80 of 2000 items from the new procedure were found to be defective, find 90% confidence interval for the true difference in the fraction of defectives between the existing and the new process.

Solution:

Sample 1

$$n_1 = 1500 \quad x_1 = 75$$

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{75}{1500} = 0.05$$

$$\hat{q}_1 = 1 - \hat{p}_1 = 1 - 0.05 = 0.95$$

Sample 2

$$n_2 = 2000 \quad x_2 = 80$$

$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{80}{2000} = 0.04$$

$$\hat{q}_2 = 1 - \hat{p}_2 = 1 - 0.04 = 0.96$$

90% C.I. for $p_1 - p_2$

$$1 - \alpha = 0.90 \qquad \alpha = 0.1$$

$$\frac{\alpha}{2} = 0.05 \qquad 1 - \frac{\alpha}{2} = 0.95$$

$$Z_{0.05} = 1.645$$

Z	0.04	0.05
1.6	0.9495	0.9505

$$(\hat{P}_1 - \hat{P}_2) \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \ \hat{q}_1}{n_1} + \frac{\hat{p}_2 \ \hat{q}_2}{n_2}}$$

$$(0.05 - 0.04) \pm 1.645 \sqrt{\frac{0.05 \times 0.95}{1500} + \frac{0.04 \times 0.96}{2000}}$$

[0.0017 , 0.0217]

Chapter 3 : Test of hypothesis

عند إدعاء باحث إدعاء معين فكلامه قد يكون صحيحاً أو خطأ
فمثلاً لو إدعى شخص أن متوسط الطول عند المصريين هو cm ١.٨٠ فكلامه قد يحتمل الصواب وقد يحتمل الخطأ

$\mu = 1.80$ وممكن $\mu \neq 1.80$, $\mu > 1.80$, $\mu < 1.80$

فيوجد نوعين لكل فرض أو إدعاء

الفرض الصفري
Null hypothesis
$H_o: \mu = 1.80$

الفرض البديل
Alternative hypothesis
$H_1: \mu \neq 1.80$
$\mu > 1.80$
$\mu < 1.80$

⇨ test of hypothesis about population mean

Step 1 : state null and alternative hypothesis

$H_o: \mu = \mu_o$ v.s. $H_1: \mu \neq \mu_o$ two sided alternative hypothesis

$H_1: \mu > \mu_o$ one sided alternative hypothesis

$H_1: \mu < \mu_o$ one sided alternative hypothesis

Step 2 : determine the distribution and test statistics

σ

Known

$$Z = \frac{\bar{X} - \mu_o}{\sigma / \sqrt{n}}$$

Normal distribution

Unknown

n

n ≥ 30

$$Z = \frac{\bar{X} - \mu_o}{S / \sqrt{n}}$$

Approximate Normal distribution

n < 30

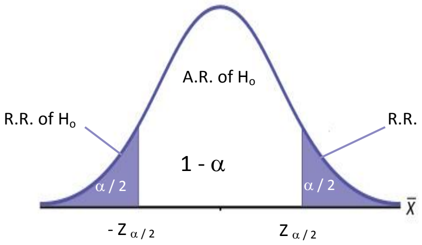
$$T = \frac{\bar{X} - \mu_o}{S / \sqrt{n}}$$

T- distribution

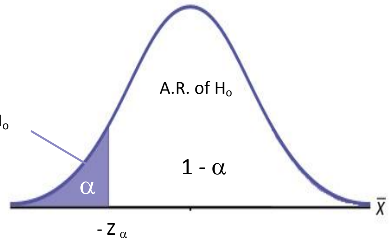
Step 3 : determine significance level α

Step 4 : determine critical region

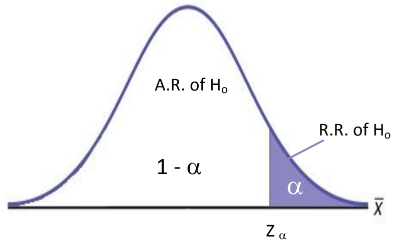
H_1 — Z, T



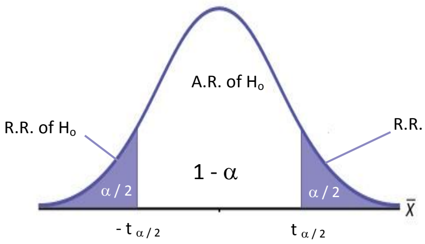
$H_1 : \mu \neq \mu_o$



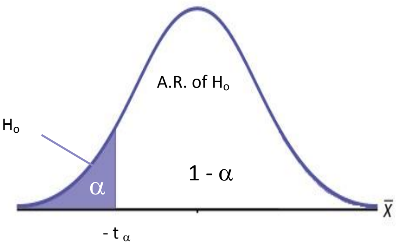
$H_1 : \mu < \mu_o$



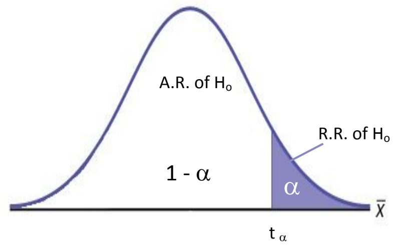
$H_1 : \mu > \mu_o$



$H_1 : \mu \neq \mu_o$



$H_1 : \mu < \mu_o$



$H_1 : \mu > \mu_o$

Accepted region of H_o منطقة القبول

rejected region of H_o منطقة الرفض

Step 5 : make a decision

if the calculate statistics lie in accepted region of Ho ,then accept the Ho and reject H1

Example 7.2 [page 119]

A random sample of 100 recorded deaths in the United States during the past year showed an average of 71.8 years. Assuming a population standard deviation of 8.9 year, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

Solution:

$$n = 100 \quad \bar{X} = 71.8 \quad \sigma = 8.9$$

step 1 : state null and alternative hypothesis

$H_o : \mu = 70$ v.s. $H_1 : \mu > 70$

ممكن تبقى عكس الفرض البديل ≤

step 2 : determine the distribution and test statistics

since σ is known

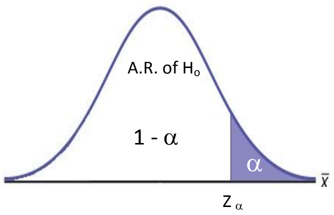
we use the normal distribution

$$Z = \frac{\bar{X} - \mu_o}{\sigma / \sqrt{n}} = \frac{71.8 - 70}{8.9 / \sqrt{100}} = 2.02$$

step 3 : determine significance level α

$\alpha = 0.05$

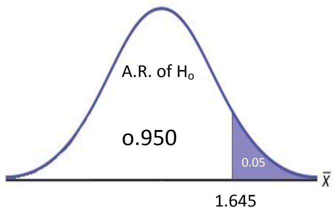
step 4 : determine critical region



α = 0.05

1 - α = 0.950

$Z\alpha$ = 1.645



step 5 : make a decision

since Z = 2.02 lie in rejected region of H_o
reject H_o and accept H_1
$\mu > 70$

Example 7.3 [page 120]

A manufacturer of sports equipment has developed a new synthetic fishing line that he claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that $\mu = 8$ kg against the alternative that $\mu \neq 8$ kg if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kg. Use $\alpha = 0.01$.

Solution:

$$n = 50 \quad \bar{X} = 7.8 \quad \sigma = 0.5$$

step 1 : state null and alternative hypothesis

H_o : $\mu = 8$ v.s. H_1 : $\mu \neq 8$

step 2 : determine the distribution and test statistics

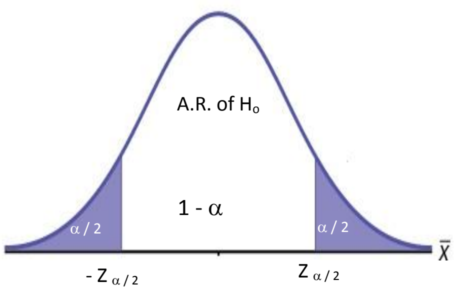
since σ is known
we use the normal distribution

$$Z = \frac{\bar{X} - \mu_o}{\sigma / \sqrt{n}} = \frac{7.8 - 8}{0.5 / \sqrt{50}} = -2.83$$

step 3 : determine significance level α

α = 0.01

step 4 : determine critical region

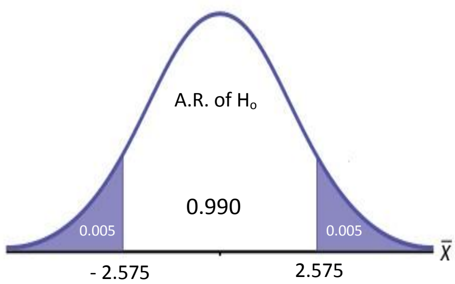


α = 0.01

$\frac{\alpha}{2}$ = 0.005

1 - $\frac{\alpha}{2}$ = 0.995

$Z_{\alpha/2}$ = 2.575



step 5 : make a decision

since Z = 2.575 lie in rejected region of H_o
reject H_o and accept H_1
$\mu \neq 8$

Example 7.4 [page 121]

If a random sample of 12 homes included in a planned study indicates that vacuum cleaners expend an average of 42 kilowatt-hours per year with a standard deviation of 11.9 kilowatt-hours, does this suggest at the 0.05 level of significance that the vacuum cleaners expend, on the average, less than 46 kilowatt-hours annually? Assume the population of kilowatt-hours to be normal.

Solution:

$$n = 12 \quad \bar{X} = 42 \quad S = 11.9$$

step 1 : state null and alternative hypothesis

$H_o: \mu = 46$ v.s. $H_1: \mu < 8$

step 2 : determine the distribution and test statistics

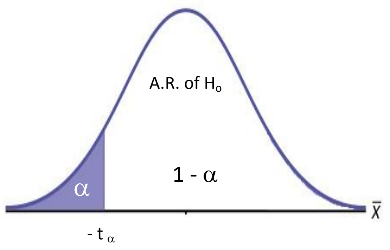
since σ is unknown and n = 12 < 30
we use the T- distribution

$$T = \frac{\bar{X} - \mu_o}{S / \sqrt{n}} = \frac{42 - 46}{11.9 / \sqrt{12}} = -1.16$$

step 3 : determine significance level α

$\alpha = 0.05$

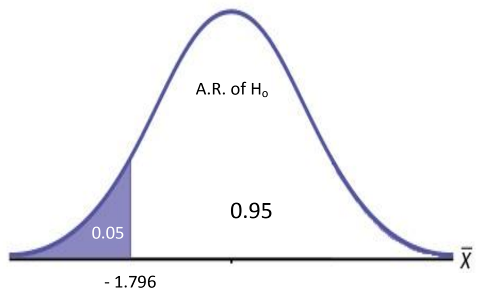
step 4 : determine critical region



$\alpha = 0.05$

$df = n - 1 = 11$

$t_{\alpha, df} = 1.796$



step 5 : make a decision

accepted region of H_o

⇨ test of hypothesis about the difference between difference two mean

عند إدعاء باحث إدعاء معين فكلامه قد يكون صحيحاً أو خطأ
فمثلاً لو إدعى شخص أن الفرق بين متوسط الطول في مصر والسودان هو X cm فكلامه قد يحتمل الصواب وقد يحتمل الخطأ
وللتأكد من ذلك نتبع الخطوات التاليه

Step 1 : state null and alternative hypothesis

$H_o: \mu_1 = \mu_2$ v.s. $H_1: \mu_1 \neq \mu_2$ two sided alternative hypothesis

$H_1: \mu_1 > \mu_2$ one sided alternative hypothesis

$H_1: \mu_1 < \mu_2$ one sided alternative hypothesis

Step 2 : determine the distribution and test statistics

If σ_1 , σ_2

Known

Use the normal distribution

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Unknown

n_1 , n_2

n ≥ 30

Use the approximate normal distribution

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

n < 30

Use the t-distribution

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

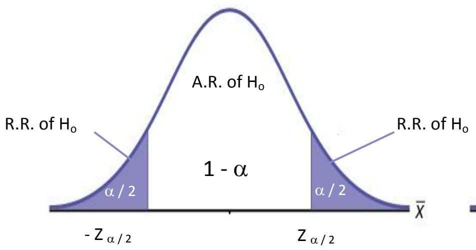
$$Sp = \sqrt{\frac{(n_1 - 1)\, S_1^2 + (n_2 - 1)\, S_2^2}{n_1 + n_2 - 2}}$$

$$df = n_1 + n_2 - 2$$

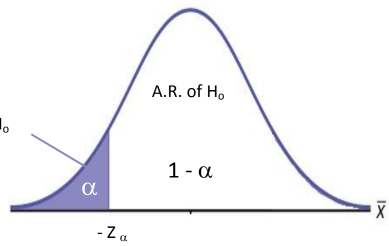
Step 3 : determine significance level α

Step 4 : determine critical region of H_o

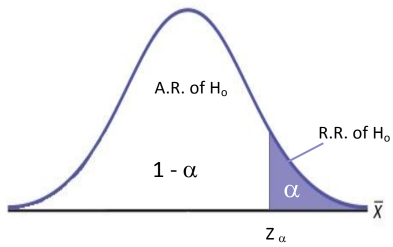
$$H_1 \begin{cases} Z \\ T \end{cases}$$



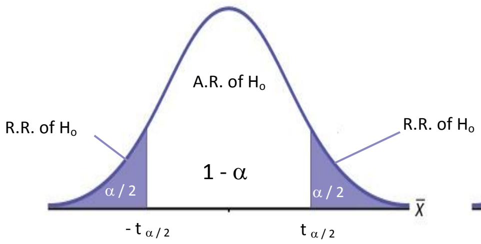
$H_1 : \mu \neq \mu_o$



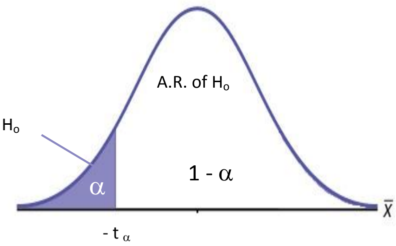
$H_1 : \mu < \mu_o$



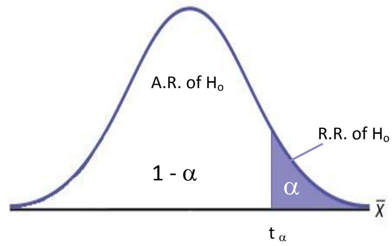
$H_1 : \mu > \mu_o$



$H_1 : \mu \neq \mu_o$



$H_1 : \mu < \mu_o$



$H_1 : \mu > \mu_o$

☐ Accepted region of H_o منطقة القبول

■ rejected region of H_o منطقة الرفض

Step 5 : make a decision

If Z or t lie in A.R. of H_o then accept H_o and region H_1

If Z or t lie in R.R. of H_o then accept H_1 and region H_o

Example 7.7 [page 125]

To compare customer satisfaction levels of two competing cable television companies, 174 customers of Company 1 and 355 customers of Company 2 were randomly selected and were asked to rate their cable companies on a five-point scale, with 1 being least satisfied and 5 most satisfied. The survey results are summarized in the following table:
Company 1: $n_1=174$, $\overline{X}_1=3.51$, $S_1=0.51$
and Company 2: $n_2=355$, $\overline{X}_2=3.24$, $S_2=0.52$
Test at the 1% level of significance whether the data provide sufficient evidence to conclude that Company 1 has a higher mean satisfaction rating than does Company 2. Use the critical value approach.

Solution:

$$n_1 = 174 \quad \overline{X}_1 = 3.51 \quad S_1 = 0.51$$

$$n_2 = 355 \quad \overline{X}_2 = 3.24 \quad S_2 = 0.52$$

step 1 : state null and alternative hypothesis

$H_o : \mu_1 = \mu_2$ v.s. $H_1 : \mu_1 > \mu_2$

$\mu_1 - \mu_2 = 0$ $\mu_1 - \mu_2 > 0$

step 2 : determine the distribution and test statistics

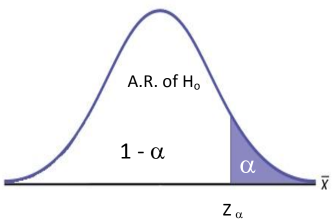
since σ_1 , σ_2 is unknown and n_1 , $n_2 > 30$ then
we use the approximate normal distribution

$$Z = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{(3.51 - 3.24) - (0)}{\sqrt{\frac{0.51^2}{174} + \frac{0.52^2}{355}}} = 5.684$$

step 3 : determine significance level α

$\alpha = 0.01$

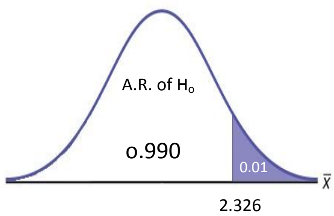
step 4 : determine critical region



$\alpha = 0.01$

$1 - \alpha = 0.990$

$Z\alpha = 2.326$



step 5 : make a decision

since $Z = 5.684$ lie in rejected region of H_o
reject H_o and accept H_1
$\mu_1 > \mu_2$

Example 7.9 [page 126]

An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average wear of 85 units with a sample standard deviation of 4, while the samples of materials 2 gave an average wear of 81 and a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the mean abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume populations to be approximately normal with equal variances.

Solution:

$$n_1 = 12 \quad \bar{X}_1 = 85 \quad S_1 = 4$$

$$n_2 = 10 \quad \bar{X}_2 = 81 \quad S_2 = 5$$

step 1 : state null and alternative hypothesis

$H_o : \mu_1 = \mu_2 + 2$ v.s. $H_1 : \mu_1 > \mu_2 + 2$

$\mu_1 - \mu_2 = 2$ $\mu_1 - \mu_2 > 2$

step 2 : determine the distribution and test statistics

since σ_1 , σ_2 is unknown and n_1 , $n_2 < 30$ then
we use the t-distribution

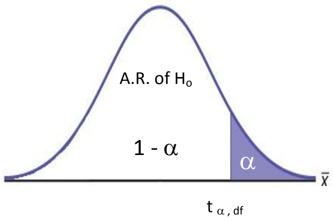
$$Sp = \sqrt{\frac{(n_1 - 1)\ S_1^2 \ + \ (n_2 - 1)\ S_2^2}{n_1 \ + \ n_2 \ - 2}} = \sqrt{\frac{(12 - 1)\ 4^2 \ + \ (10 - 1)\ 5^2}{12 \ + \ 10 \ - 2}} = 4.478$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{Sp\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(85 - 81) - (2)}{4.478\sqrt{\frac{1}{12} + \frac{1}{10}}} = 1.04$$

step 3 : determine significance level α

$\alpha = 0.05$

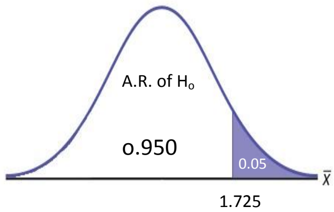
step 4 : determine critical region



$\alpha = 0.05$

$df = n_1 + n_2 - 2 = 20$

$t_{\alpha\ ,\ df} = 1.725$



step 5 : make a decision

since t = 1.04 lie in accepted region of H_o
accept H_o and reject H_1
$\mu_1 = \mu_2 + 2$

⇨ test of hypothesis about population proportion

عند إدعاء باحث إدعاء معين فكلامه قد يكون صحيحاً أو خطأ
فمثلاً لو إدعى شخص أن نسبة الطوال في مصر هي X فكلامه قد يحتمل الصواب وقد يحتمل الخطأ
وللتأكد من ذلك نتبع الخطوات التاليه

Step 1 : state null and alternative hypothesis

$H_0 : p = p_o$ v.s. $H_1 : p \neq p_o$ two sided alternative hypothesis

$q_o = 1 - p_o$ $H_1 : p > p_o$ one sided alternative hypothesis

$H_1 : p < p_o$ one sided alternative hypothesis

Step 2 : determine the distribution and test statistics

If $n\hat{p} > 5$ and $n\hat{q} > 5$

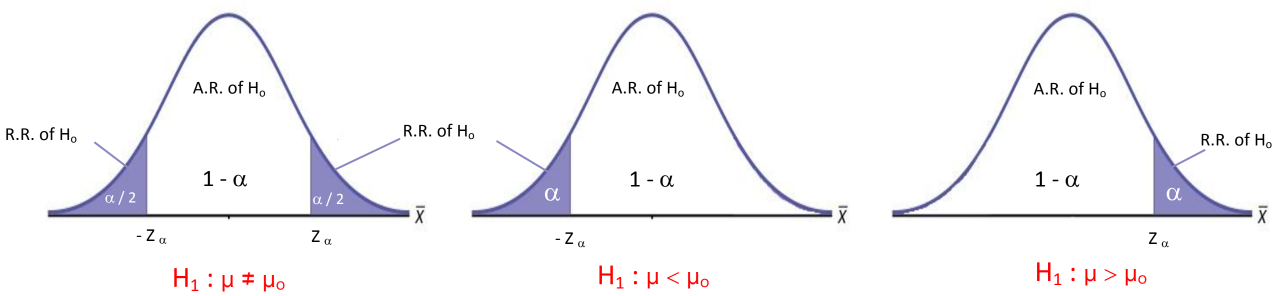
, then we use the normal distribution

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p_o q_o}{n}}}$$

$n\hat{p} > 5$ and $n\hat{q} > 5$
العينه بتلك المواصفات تسمى
Large sample
وهي شرط هام لإستخدام
التوزيع الطبيعي

Step 3 : determine significance level α

Step 4 : determine critical region of H_o



Accepted region of H_o منطقة القبول

rejected region of H_o منطقة الرفض

Step 5 : make a decision

If Z or t lie in A.R. of H_o then accept H_o and region H_1

If Z or t lie in R.R. of H_o then accept H_1 and region H_o

Example 7.10 [page 128]

A builder claims that heat pumps are installed in 70% of all homes being constructed today in the city of Cairo. Would you agree with this claim if a random survey of new homes in the city shows that 8 out of 15 homes had heat pumps installed? Use a 0.10 level of significance.

Solution:

$$n = 15 \quad p_o = 0.70 \quad q_o = \; 1 - p_o = 0.30$$

$$X = 8 \quad \hat{p} = \frac{X}{n} \; = \; \frac{8}{10} \; = 0.533$$

step 1 : state null and alternative hypothesis

$H_o : \; p \; = \; 0.70$ v.s. $H_1 : \; p \; \neq \; 0.70$

step 2 : determine the distribution and test statistics

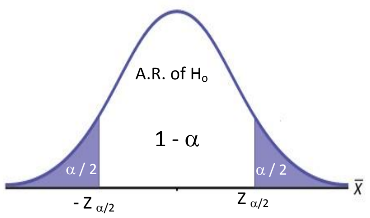
since $n\hat{p} = \; 7.02 \; > \; 5$ and $n\hat{q} \; = \; 7.01 \; > \; 5$
, then we use the normal distribution

$$Z = \frac{\hat{p} \; - \; p}{\sqrt{\frac{p_o \, q_o}{n}}} \; = \; \frac{0.533 \; - \; 0.7}{\sqrt{\frac{0.70 \; \times \; 0.30}{10}}} \; = \; - \, 1.41$$

step 3 : determine significance level α

α = 0.1

step 4 : determine critical region

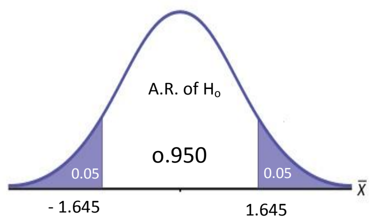


α = 0.1

$$\frac{\alpha}{2} = 0.050$$

$$1 - \frac{\alpha}{2} \; = \; 0.950$$

$Z_{\alpha / 2}$ = 1.645



step 5 : make a decision

since Z = 1.645 lie in accepted region of H_o
accept H_o and reject H_1
$p \; = \; 0.70$

⇨ Test of hypothesis about the difference between two population proportion

عند إدعاء باحث إدعاء معين فكلامه قد يكون صحيحاً أو خطأ

فمثلاً لو إدعى شخص أن الفرق بين نسبة الطوال في مصر ونسبة الطوال في السودان هو X فكلامه قد يحتمل الصواب وقد يحتمل الخطأ

وللتأكد من ذلك نتبع الخطوات التالية

Step 1 : state null and alternative hypothesis

$H_o : p_1 = p_2$ v.s. $H_1 : p_1 \neq p_2$ two sided alternative hypothesis

$H_1 : p_1 > p_2$ one sided alternative hypothesis

$H_1 : p_1 < p_2$ one sided alternative hypothesis

Step 2 : determine the distribution and test statistics

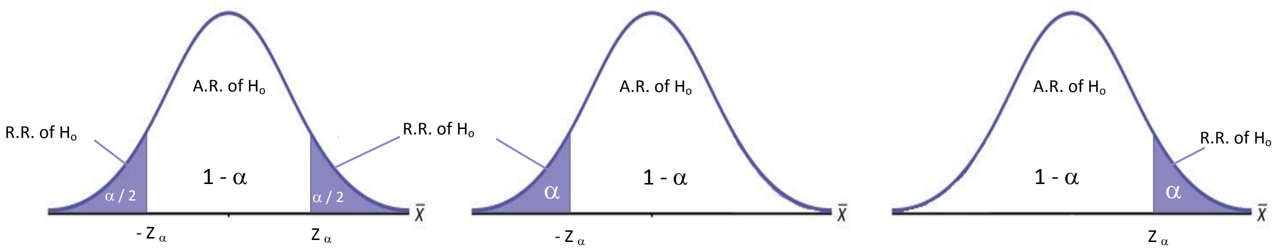
$$\text{If } n_1\hat{p}_1 > 5 \; and \; n_1\hat{q}_1 > 5 \text{ and } n_2\hat{p}_2 > 5 \; and \; n_2\hat{q}_2 > 5$$

, then we use the normal distribution

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\,(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2\,(1 - \hat{p}_2)}{n_2}}} = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\,\hat{q}_1}{n_1} + \frac{\hat{p}_2\,\hat{q}_2}{n_2}}}$$

Step 3 : determine significance level α

Step 4 : determine critical region of H_o



$H_1 : \mu \neq \mu_o$ $H_1 : \mu < \mu_o$ $H_1 : \mu > \mu_o$

Accepted region of H_o منطقة القبول

rejected region of H_o منطقة الرفض

Step 5 : make a decision

If Z or t lie in A.R. of H_o then accept H_o and region H_1

If Z or t lie in R.R. of H_o then accept H_1 and region H_o

Example 7.14 [page 132]

In the past decade intensive antismoking campaigns have been sponsored by both federal and private agencies. Suppose the American Cancer Society randomly sampled 1,500 adults in 1992 and then sampled 1,750 adults in 2002 to determine whether there was evidence that the percentage of smokers had decreased. Let $X_1 = 555$ and $X_2 = 578$ represent the numbers of smokers in the 1992 and 2002 samples, respectively. Do these data indicate that the fraction of smokers decreased over this 10-year period? Use $\alpha = 0.5$.

Solution:

$$n_1 = 1500 \quad X_1 = 550 \quad \hat{p}_1 = 0.37$$

$$n_2 = 1750 \quad X_2 = 578 \quad \hat{p}_2 = 0.33$$

step 1 : state null and alternative hypothesis

$H_o: p_1 = p_2$ v.s. $H_1: p_1 > p_2$

$p_1 - p_2 = 0$ $p_1 - p_2 > 0$

step 2 : determine the distribution and test statistics

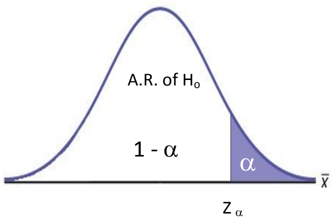
since $n_1\hat{p}_1 > 5$ and $n_1\hat{q}_1 > 5$ and $n_2\hat{p}_2 > 5$ and $n_2\hat{q}_2 > 5$
, then we use the normal distribution

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} = \frac{(0.37 - 0.33) - (0)}{\sqrt{\frac{0.37(1-0.37)}{1500} + \frac{0.33(1-0.33)}{1750}}} = 0.237$$

step 3 : determine significance level α

$\alpha = 0.05$

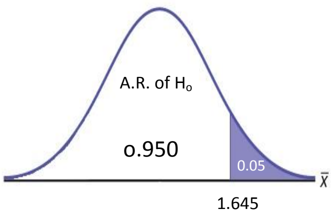
step 4 : determine critical region



$\alpha = 0.05$

$1 - \alpha = 0.950$

$Z_\alpha = 1.645$



step 5 : make a decision

since Z = 0.237 lie in accepted region of H_o
accept H_o and reject H_1
$p_1 = p_2$

Chapter 4 : Analysis of variance [ANOVA]

في حالة وجود أكثر من مجتمعين وأدعى شخص مثلاً ان متوسطات تلك المجتمعات متساويه
فلتحديد صحة كلامه لا نستطيع إستخدام الطريقة السابقه **Test of Hypothesis** فهي تستخدم فقط في حالة وجود مجتمعين بحد أقصى
ونستخدم بدلاً منها طريقه أخرى تسمى **ANOVA**

μ_1 μ_2

$H_o : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$

μ_1 μ_2 μ_3 .... μ_{12}

عدد الإختبارات (Test of Hypothesis) اللازمه لحساب الفرق بين الميلين والتأكد من الإدعاء

$$\binom{12}{2} = 66$$

وبالتالي لا نستطيع استخدام test of hypothesis لأكثر من مجتمعين

⇨ ANOVA approach

Step 1 : state the hypothesis

$$H_o : \quad \mu_1 = \mu_2 \ \ldots\ldots\ldots\ldots\ \mu_k$$

v.s.

H_1 : at least two of the population's means not equal

Step 2 : Construct ANOVA table :

Source of variation	Sum of squares مجموع المربعات (SS)	Degree of freedom درجات الحريه (df)	Mean sum of squares متوسط مجموع الربعات (MSS)	F - computed
Treatments الإختلاف بين المجتمعات	SST_r	K - 1	$MSST_r = \frac{SST_r}{K-1}$	$MSST_r$ / MSSE
Error الإختلاف داخل كل عينه	SSE	K (n – 1)	$MSSE = \frac{SSE}{K\ (n-1)}$	
Total	SST	nk - 1		

Step 3 : determine critical region

$$f_{(K-1)\,,\,K\,(n-1)}$$

Determine from the table in the book

Step 4 : make a decision

$$\text{If } f_c > f_{(K-1)\,,\,K\,(n-1)}$$

reject H_o and accept H_1

Important laws

Correction factor (C.F.) = $\dfrac{y_{..}^2}{nk}$

SST = $\sum(\sum(y_{i.})^2) - \text{C. F.}$

SST$_r$ = $\dfrac{y_{1.}^2}{n_1} + \dfrac{y_{2.}^2}{n_2} \ldots \ldots \ldots \ldots \ldots \ldots \ldots \ldots \ldots \ldots . - C.F.$

SSE = $\text{SST} - \text{SST}_r$

K_i عدد المجتمعات
n_i حجم العينه
$Y_{i.}$ مجموع مشاهدات كل عينه
$y_{..}$ مجموع مشاهدات كل العينات

Example 8.1 [page 138]

The table below shows the lifetime under controlled conditions, in hours in excess of 1000 hours, of samples of 60 W electric light bulbs of three different brands:

Brand 1	Brand 2	Brand 3
16	18	26
15	22	31
13	20	24
21	16	30
15	24	24

Assuming all lifetimes to be normally distributed with common variance, test at the 1% significance level the hypothesis that there is no difference between the three brands with respect to mean lifetime.

Solution:

$$K = 3 \quad n = 5 \quad \alpha = 0.01$$

step 1 : state the hypothesis

H_o : $\mu_1 = \mu_2 = \mu_3$ v.s. H_1 : at least two of the population's means not equal

step 2 : Construct ANOVA table :

n	5	5	5
$Y_{i.}$ مجموع مشاهدات كل عينه	80 16 + 15 + 13 …	100 18 + 22 + 20 …	135 26 + 31 + 24 …
$Y_{i.}^2$ مربع مجموع مشاهدات كل عينه	6400 $(16 + 15 + 13 \ldots)^2$	10000 $(18 + 22 + 20 \ldots)^2$	18225 $(26 + 31 + 24 \ldots)^2$
$\sum (Y_i)^2$	1316 $16^2 + 15^2 + 13^2 \ldots$	2040 $18^2 + 22^2 + 20^2 \ldots$	3689 $26^2 + 31^2 + 24^2 \ldots$
$y_{..} = 315$			

C.F. = $\frac{y_{..}^2}{nk} = \frac{315^2}{15} = 6615$

SST = $\sum(\sum(y_{i.})^2) - \text{C.F.} = 7045 - 6615 = 430$

SST$_r$ = $\frac{y_{1.}^2}{n_1} + \frac{y_{2.}^2}{n_2} + \frac{y_{3.}^2}{n_3} - C.F. = \frac{6400 + 10000 + 18225}{5} - 6615 = 310$

SSE = $SST - SST_r = 430 - 310 = 120$

Source of variation	SS	df	MSS	F - computed
Treatments الإختلاف بين المجتمعات	310	2	155	15.5
Error الإختلاف داخل كل عينه	120	12	10	
Total	430	14		

step 3 : determine critical region

$$f_{(K-1), K(n-1)} = f_{2, 12}$$

من الجدول $f_{0.01, v1, v2}$ حيث الخطوط الأفقيه تعبر عن v_1 بينما الخطوط الرأسيه تعبر عن v_2

$$f_{2, 12} = 9.33$$

step 4 : make a decision

Since $f_c > f_{(K-1), K(n-1)}$

reject H_o and accept H_1

Example 8.2 [page 139]

Suppose the National Transportation Safety Board (NTSB) wants to examine the safety of compact cars, midsize cars, and full-size cars. It collects a sample of three for each of the treatments (car types). Using the hypothetical data provided below, test whether the mean pressure applied to the driver's head during a crash test is equal for each type of car. Use α=5%.

Compact cars	Midsize cars	Full-size cars
643	469	484
655	427	456
702	525	402

Solution:

$$K = 3 \quad n = 3 \quad \alpha = 0.05$$

step 1 : state the hypothesis

H_o : $\mu_1 = \mu_2 = \mu_3$ v.s. H_1 : at least two of the population's means not equal

step 2 : Construct ANOVA table :

n	3	3	3
$Y_{i.}$	2000	1421	1342
$Y_{i.}^2$	4000000	2019241	1800964
$\sum (Y_i)^2$	1335278	677915	603796
$y_{..} = 4763$			

C.F. = $\frac{y_{..}^2}{nk} = \frac{4763^2}{9} = 2520685.444$

SST = $\sum(\sum(y_{i.})^2) - \text{C.F.} = 2616989 - 2520685.444 = 96303.556$

SST$_r$ = $\frac{y_{1.}^2}{n_1} + \frac{y_{2.}^2}{n_2} + \frac{y_{3.}^2}{n_3} - C.F. = \frac{4000000 + 2019241 + 1800964}{3} - 2520685.444 = 86049.556$

SSE = $SST - SST_r = 94303.556 - 86049.556 = 8254$

Source of variation	SS	df	MSS	F - computed
Treatments	86049.556	2	43024.778	31.2756
Error	8254	6	1375.667	
Total	96303.556	8		

step 3 : determine critical region

$$f_{(K-1),\,K(n-1)} = f_{2,6}$$

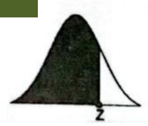
$$f_{2,6} = 5.14$$

step 4 : make a decision

$$\text{Since } f_c > f_{(K-1),\,K(n-1)}$$

reject H_o and accept H_1

Tables

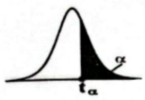


Areas Under The Standard Normal Curve

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	*0.0019*	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

t – distribution table



ν							α						
	0.40	0.30	0.20	0.15	0.10	0.05	0.025	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706	15.895	21.205	31.821	42.434	63.657	127.322	636.590
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303	4.849	5.643	6.965	8.073	9.925	14.089	31.598
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182	3.482	3.896	4.541	5.047	5.841	7.453	12.924
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447	2.612	2.829	3.143	3.372	3.707	4.317	5.959
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262	2.398	2.574	2.821	2.998	3.250	3.690	4.781
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	0.259	0.538	0.870	1.079	1.350	1.771	2.160	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145	2.264	2.415	2.624	2.771	2.977	3.326	4.140
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120	2.232	2.382	2.583	2.724	2.921	3.252	4.015
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086	2.197	2.336	2.528	2.661	2.845	3.153	3.850
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052	2.158	2.291	2.473	2.598	2.771	3.057	3.690
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045	2.150	2.282	2.462	2.586	2.756	3.038	3.659
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021	2.123	2.250	2.423	2.542	2.704	2.971	3.551
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960	2.054	2.170	2.326	2.432	2.576	2.807	3.291

f – distribution table ($f_{\alpha, v1, v2}$)

f – distribution table (at α = 0.01)

V_2	V_1																		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**15**	**20**	**25**	**30**	**40**	**60**	**120**	**∞**
1	4052.2	4999.3	5403.4	5624.3	5764.0	5859.0	5928.3	5981.1	6022.5	6055.9	6106.3	6157.3	6234.6	6286.8	6323.5	6365.9	6405.7	6450.3	6480.0
2	98.502	99.000	99.166	99.249	99.299	99.333	99.356	99.374	99.388	99.399	99.416	99.433	99.468	99.482	99.492	99.500	99.508	99.517	99.500
3	34.116	30.816	29.457	28.710	28.237	27.911	27.672	27.489	27.345	27.229	27.052	26.872	26.504	26.316	26.219	26.125	26.031	25.938	25.841
4	21.198	18.000	16.694	15.977	15.522	15.207	14.976	14.799	14.659	14.546	14.374	14.198	13.838	13.652	13.558	13.465	13.372	13.279	13.185
5	16.258	13.274	12.060	11.392	10.967	10.672	10.456	10.289	10.158	10.051	9.8883	9.7083	9.2023	9.0357	8.9524	8.8690	8.7857	8.7024	8.6190
6	13.745	10.925	9.7795	9.1483	8.7459	8.4661	8.2600	8.1017	7.9761	7.8741	7.7183	7.5422	7.0771	6.9278	6.8530	6.7782	6.7033	6.6285	6.5536
7	12.246	9.5465	8.4513	7.8467	7.4604	7.1914	6.9928	6.8401	6.7188	6.6201	6.4691	6.2980	5.8712	5.7347	5.6664	5.5980	5.5297	5.4613	5.3929
8	11.259	8.6491	7.5910	7.0061	6.6318	6.3707	6.1776	6.0289	5.9106	5.8143	5.6672	5.5000	5.1149	4.9872	4.9233	4.8593	4.7954	4.7314	4.6674
9	10.561	8.0215	6.9919	6.4221	6.0569	5.8018	5.6129	5.4671	5.3511	5.2568	5.1137	4.9491	4.5841	4.4631	4.4025	4.3419	4.2812	4.2205	4.1598
10	10.044	7.5594	6.5523	5.9943	5.6363	5.3858	5.2001	5.0567	4.9424	4.8496	4.7088	4.5471	4.1985	4.0830	4.0251	3.9673	3.9094	3.8515	3.7936
11	9.6460	7.2057	6.2167	5.6683	5.3160	5.0692	4.8860	4.7445	4.6320	4.5405	4.3974	4.2386	3.9036	3.7912	3.7350	3.6789	3.6228	3.5666	3.5105
12	9.3302	6.9266	5.9525	5.4120	5.0643	4.8206	4.6400	4.4997	4.3881	4.2974	4.1552	3.9985	3.6736	3.5633	3.5082	3.4530	3.3979	3.3427	3.2875
13	9.0738	6.7010	5.7394	5.2053	4.8616	4.6203	4.4410	4.3021	4.1914	4.1015	3.9605	3.8056	3.4894	3.3808	3.3265	3.2722	3.2178	3.1635	3.1091
14	8.8616	6.5149	5.5639	5.0354	4.6950	4.4558	4.2780	4.1400	4.0300	3.9409	3.8002	3.6469	3.3386	3.2313	3.1776	3.1240	3.0703	3.0166	2.9629
15	8.6831	6.3589	5.4170	4.8932	4.5556	4.3180	4.1415	4.0045	3.8951	3.8060	3.6652	3.5133	3.2119	3.1057	3.0526	2.9995	2.9464	2.8933	2.8402
16	8.5310	6.2262	5.2922	4.7726	4.4374	4.2016	4.0261	3.8900	3.7811	3.6925	3.5496	3.3988	3.1035	2.9982	2.9456	2.8931	2.8405	2.7879	2.7353
17	8.3997	6.1121	5.1850	4.6690	4.3359	4.1016	3.9270	3.7915	3.6832	3.5951	3.4493	3.2998	3.0098	2.9053	2.8532	2.8011	2.7490	2.6969	2.6447
18	8.2854	6.0129	5.0919	4.5790	4.2479	4.0148	3.8411	3.7062	3.5984	3.5106	3.3616	3.2134	2.9280	2.8243	2.7726	2.7209	2.6692	2.6175	2.5657
19	8.1849	5.9259	5.0103	4.5003	4.1708	3.9390	3.7659	3.6316	3.5242	3.4367	3.2843	3.1373	2.8559	2.7529	2.7016	2.6503	2.5989	2.5476	2.4962
20	8.0960	5.8489	4.9382	4.4307	4.1030	3.8714	3.6987	3.5644	3.4571	3.3699	3.2157	3.0697	2.7918	2.6896	2.6387	2.5878	2.5368	2.4859	2.4349
21	8.0166	5.7804	4.8740	4.3688	4.0425	3.8118	3.6397	3.5059	3.3989	3.3119	3.1542	3.0092	2.7344	2.6329	2.5823	2.5317	2.4811	2.4305	2.3799
22	7.9454	5.7190	4.8172	4.3143	3.9890	3.7590	3.5875	3.4540	3.3474	3.2607	3.0989	2.9548	2.6828	2.5819	2.5316	2.4813	2.4310	2.3807	2.3304
23	7.8811	5.6637	4.7659	4.2650	3.9407	3.7115	3.5405	3.4074	3.3011	3.2147	3.0488	2.9056	2.6361	2.5357	2.4857	2.4357	2.3857	2.3357	2.2857
24	7.8229	5.6136	4.7195	4.2204	3.8970	3.6686	3.4980	3.3653	3.2592	3.1730	3.0032	2.8609	2.5936	2.4938	2.4440	2.3942	2.3445	2.2947	2.2449
25	7.7698	5.5680	4.6775	4.1800	3.8574	3.6298	3.4596	3.3272	3.2213	3.1353	2.9613	2.8198	2.5546	2.4552	2.4057	2.3561	2.3066	2.2570	2.2074
26	7.7213	5.5264	4.6392	4.1430	3.8212	3.5942	3.4243	3.2923	3.1866	3.1008	2.9228	2.7821	2.5187	2.4197	2.3704	2.3211	2.2718	2.2225	2.1731
27	7.6767	5.4884	4.6040	4.1090	3.7879	3.5615	3.3920	3.2603	3.1548	3.0692	2.8872	2.7472	2.4855	2.3869	2.3378	2.2887	2.2397	2.1906	2.1415
28	7.6356	5.4536	4.5716	4.0777	3.7573	3.5314	3.3622	3.2308	3.1255	3.0400	2.8542	2.7149	2.4547	2.3565	2.3076	2.2587	2.2099	2.1610	2.1121
29	7.5975	5.4215	4.5417	4.0489	3.7291	3.5037	3.3348	3.2037	3.0986	3.0132	2.8235	2.6848	2.4261	2.3282	2.2795	2.2308	2.1821	2.1334	2.0847
30	7.5625	5.3903	4.5132	4.0220	3.7029	3.4779	3.3093	3.1785	3.0736	2.9883	2.7949	2.6567	2.3993	2.3017	2.2532	2.2047	2.1562	2.1076	2.0591
40	7.3141	5.1785	4.3126	3.8283	3.5138	3.2910	3.1240	2.9942	2.8906	2.8060	2.5774	2.4430	2.1938	2.0990	2.0516	2.0043	1.9569	1.9095	1.8621
60	7.0771	4.9774	4.1259	3.6491	3.3393	3.1194	2.9542	2.8258	2.7232	2.6394	2.3793	2.2480	2.0053	1.9126	1.8662	1.8199	1.7735	1.7271	1.6806
120	6.8510	4.7865	3.9498	3.4808	3.1765	2.9598	2.7962	2.6693	2.5678	2.4848	2.1958	2.0677	1.8315	1.7411	1.6957	1.6503	1.6048	1.5593	1.5138
∞	6.6349	4.6052	3.7816	3.3192	3.0173	2.8020	2.6398	2.5140	2.4135	2.3309	2.0262	1.9013	1.6710	1.5828	1.5383	1.4938	1.4493	1.4048	1.3603

f – distribution table (at α = 0.025)

V_2									V_1									
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**15**	**20**	**25**	**30**	**40**	**60**	**120**	**∞**
1	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63	976.71	984.87	993.10	998.08	1001.4	1005.6	1009.8	1014.0	1018.3
2	38.506	39.000	39.165	39.248	39.298	39.331	39.355	39.373	39.387	39.398	39.415	39.431	39.454	39.468	39.477	39.486	39.496	39.506	39.496
3	17.443	16.044	15.439	15.101	14.885	14.735	14.624	14.540	14.473	14.419	14.337	14.253	14.167	14.115	14.081	14.046	14.011	13.976	13.939
4	12.218	10.649	9.9792	9.6045	9.3645	9.1973	9.0741	8.9796	8.9047	8.8439	8.7512	8.6565	8.5599	8.5026	8.4655	8.4277	8.3899	8.3520	8.3139
5	10.007	8.4336	7.7636	7.3879	7.1464	6.9777	6.8531	6.7572	6.6810	6.6192	6.5245	6.4277	6.3286	6.2696	6.2314	6.1924	6.1534	6.1144	6.0750
6	8.8131	7.2599	6.5988	6.2272	5.9876	5.8198	5.6955	5.5996	5.5234	5.4613	5.3662	5.2688	5.1686	5.1086	5.0698	5.0302	4.9905	4.9506	4.9104
7	8.0727	6.5415	5.8898	5.5226	5.2852	5.1186	4.9949	4.8993	4.8232	4.7611	4.6660	4.5683	4.4675	4.4069	4.3676	4.3274	4.2872	4.2468	4.2062
8	7.5709	6.0595	5.4160	5.0526	4.8173	4.6517	4.5286	4.4333	4.3572	4.2951	4.1997	4.1015	4.0000	3.9390	3.8992	3.8586	3.8179	3.7770	3.7359
9	7.2093	5.7147	5.0781	4.7181	4.4844	4.3197	4.1970	4.1020	4.0260	3.9639	3.8682	3.7694	3.6672	3.6058	3.5657	3.5247	3.4836	3.4423	3.4008
10	6.9367	5.4564	4.8256	4.4683	4.2361	4.0721	3.9498	3.8549	3.7790	3.7168	3.6205	3.5211	3.4181	3.3563	3.3158	3.2743	3.2327	3.1908	3.1487
11	6.7241	5.2559	4.6300	4.2751	4.0440	3.8806	3.7586	3.6638	3.5879	3.5257	3.4290	3.3290	3.2253	3.1631	3.1222	3.0803	3.0382	2.9959	2.9534
12	6.5538	5.0959	4.4742	4.1212	3.8911	3.7283	3.6065	3.5118	3.4358	3.3736	3.2766	3.1760	3.0716	3.0089	2.9676	2.9253	2.8828	2.8400	2.7970
13	6.4143	4.9653	4.3472	3.9961	3.7667	3.6043	3.4827	3.3880	3.3120	3.2497	3.1522	3.0510	2.9459	2.8828	2.8412	2.7985	2.7556	2.7124	2.6690
14	6.2979	4.8567	4.2417	3.8922	3.6634	3.5014	3.3799	3.2853	3.2093	3.1469	3.0490	2.9472	2.8414	2.7778	2.7359	2.6929	2.6496	2.6060	2.5622
15	6.1995	4.7650	4.1528	3.8043	3.5761	3.4143	3.2927	3.1980	3.1220	3.0595	2.9612	2.8588	2.7524	2.6884	2.6462	2.6028	2.5592	2.5152	2.4710
16	6.1151	4.6867	4.0768	3.7294	3.5017	3.3402	3.2187	3.1240	3.0479	2.9853	2.8866	2.7837	2.6767	2.6123	2.5698	2.5261	2.4821	2.4378	2.3932
17	6.0420	4.6189	4.0112	3.6648	3.4375	3.2761	3.1546	3.0599	2.9838	2.9211	2.8221	2.7188	2.6113	2.5465	2.5037	2.4598	2.4155	2.3709	2.3260
18	5.9781	4.5597	3.9539	3.6083	3.3814	3.2201	3.0986	3.0039	2.9277	2.8650	2.7657	2.6620	2.5541	2.4889	2.4459	2.4017	2.3571	2.3122	2.2670
19	5.9216	4.5075	3.9034	3.5587	3.3322	3.1710	3.0496	2.9548	2.8786	2.8158	2.7162	2.6122	2.5039	2.4384	2.3952	2.3508	2.3059	2.2608	2.2153
20	5.8715	4.4613	3.8587	3.5147	3.2885	3.1274	3.0061	2.9113	2.8351	2.7723	2.6725	2.5682	2.4596	2.3938	2.3504	2.3057	2.2606	2.2151	2.1693
21	5.8266	4.4199	3.8188	3.4754	3.2494	3.0885	2.9672	2.8724	2.7962	2.7333	2.6333	2.5287	2.4198	2.3537	2.3101	2.2652	2.2198	2.1741	2.1280
22	5.7863	4.3828	3.7829	3.4401	3.2144	3.0536	2.9324	2.8376	2.7614	2.6985	2.5983	2.4935	2.3843	2.3180	2.2742	2.2291	2.1835	2.1375	2.0911
23	5.7498	4.3492	3.7505	3.4081	3.1826	3.0220	2.9008	2.8060	2.7298	2.6669	2.5665	2.4615	2.3521	2.2855	2.2416	2.1963	2.1505	2.1042	2.0576
24	5.7166	4.3188	3.7211	3.3790	3.1538	2.9932	2.8721	2.7773	2.7011	2.6382	2.5377	2.4325	2.3229	2.2561	2.2120	2.1666	2.1206	2.0741	2.0272
25	5.6864	4.2911	3.6943	3.3525	3.1274	2.9670	2.8459	2.7512	2.6750	2.6121	2.5115	2.4061	2.2963	2.2294	2.1852	2.1396	2.0934	2.0467	1.9996
26	5.6586	4.2657	3.6697	3.3282	3.1034	2.9430	2.8220	2.7273	2.6511	2.5882	2.4875	2.3820	2.2721	2.2050	2.1606	2.1148	2.0685	2.0216	1.9743
27	5.6331	4.2424	3.6472	3.3059	3.0812	2.9209	2.8000	2.7053	2.6291	2.5662	2.4655	2.3599	2.2499	2.1826	2.1382	2.0923	2.0458	1.9987	1.9512
28	5.6096	4.2209	3.6264	3.2853	3.0607	2.9005	2.7796	2.6849	2.6088	2.5459	2.4451	2.3394	2.2292	2.1618	2.1173	2.0713	2.0246	1.9773	1.9296
29	5.5878	4.2009	3.6072	3.2662	3.0418	2.8817	2.7609	2.6662	2.5901	2.5272	2.4264	2.3205	2.2103	2.1428	2.0981	2.0520	2.0051	1.9577	1.9097
30	5.5675	4.1821	3.5894	3.2486	3.0243	2.8643	2.7435	2.6489	2.5727	2.5098	2.4090	2.3030	2.1926	2.1250	2.0803	2.0341	1.9871	1.9395	1.8914
40	5.4239	4.0510	3.4633	3.1252	2.9027	2.7435	2.6231	2.5286	2.4525	2.3896	2.2886	2.1818	2.0702	2.0016	1.9563	1.9094	1.8616	1.8130	1.7637
60	5.2856	3.9253	3.3425	3.0069	2.7858	2.6274	2.5073	2.4130	2.3370	2.2742	2.1731	2.0658	1.9531	1.8838	1.8379	1.7902	1.7415	1.6919	1.6415
120	5.1523	3.8043	3.2260	2.8926	2.6730	2.5152	2.3952	2.3010	2.2250	2.1622	2.0609	1.9531	1.8395	1.7694	1.7228	1.6743	1.6246	1.5739	1.5222
∞	5.0239	3.6889	3.1142	2.7829	2.5646	2.4076	2.2878	2.1936	2.1176	2.0548	1.9531	1.8446	1.7300	1.6589	1.6116	1.5622	1.5114	1.4593	1.406

f – distribution table (at $\alpha = 0.05$)

V_2	V_1																		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**12**	**15**	**20**	**25**	**30**	**40**	**60**	**120**	**∞**
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.26	250.10	251.14	252.20	253.25	254.31
2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385	19.396	19.413	19.429	19.446	19.456	19.462	19.471	19.479	19.487	19.496
3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855	8.7446	8.7029	8.6602	8.6344	8.6173	8.5984	8.5790	8.5595	8.5399
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	5.9988	5.9644	5.9117	5.8578	5.8025	5.7705	5.7495	5.7270	5.7043	5.6814	5.6586
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351	4.6777	4.6188	4.5581	4.5228	4.5005	4.4761	4.4513	4.4264	4.4014
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2066	4.1468	4.0990	4.0600	4.0000	3.9381	3.8742	3.8367	3.8130	3.7870	3.7606	3.7340	3.7072
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767	3.6365	3.5747	3.5107	3.4445	3.4055	3.3807	3.3538	3.3263	3.2987	3.2709
8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472	3.2839	3.2184	3.1503	3.1103	3.0847	3.0571	3.0289	3.0005	2.9719
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373	3.0729	3.0061	2.9365	2.8957	2.8694	2.8411	2.8121	2.7829	2.7534
10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782	2.9130	2.8450	2.7740	2.7324	2.7055	2.6766	2.6470	2.6171	2.5870
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962	2.8536	2.7876	2.7186	2.6464	2.6041	2.5768	2.5473	2.5170	2.4865	2.4556
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534	2.6866	2.6169	2.5436	2.5005	2.4727	2.4427	2.4119	2.3807	2.3492
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.6710	2.6037	2.5331	2.4589	2.4152	2.3870	2.3565	2.3251	2.2933	2.2612
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022	2.5342	2.4630	2.3879	2.3435	2.3149	2.2840	2.2522	2.2199	2.1872
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437	2.4753	2.4035	2.3275	2.2826	2.2537	2.2223	2.1900	2.1572	2.1240
16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935	2.4247	2.3524	2.2756	2.2302	2.2009	2.1692	2.1365	2.1033	2.0696
17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943	2.4499	2.3807	2.3079	2.2304	2.1846	2.1550	2.1230	2.0899	2.0562	2.0221
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117	2.3421	2.2689	2.1906	2.1444	2.1145	2.0822	2.0487	2.0147	1.9801
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779	2.3080	2.2343	2.1555	2.1089	2.0787	2.0461	2.0123	1.9779	1.9429
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928	2.3479	2.2776	2.2036	2.1242	2.0772	2.0467	2.0138	1.9796	1.9448	1.9094
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660	2.3209	2.2504	2.1760	2.0960	2.0487	2.0180	1.9848	1.9503	1.9151	1.8793
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967	2.2258	2.1511	2.0707	2.0230	1.9921	1.9586	1.9238	1.8882	1.8520
23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201	2.2747	2.2036	2.1286	2.0476	1.9997	1.9685	1.9348	1.8997	1.8638	1.8272
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547	2.1834	2.1080	2.0267	1.9784	1.9470	1.9130	1.8776	1.8413	1.8044
25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821	2.2365	2.1649	2.0893	2.0075	1.9590	1.9274	1.8931	1.8574	1.8208	1.7835
26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197	2.1479	2.0720	1.9898	1.9411	1.9093	1.8748	1.8388	1.8019	1.7642
27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501	2.2043	2.1323	2.0561	1.9736	1.9246	1.8926	1.8579	1.8216	1.7844	1.7464
28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360	2.1900	2.1179	2.0415	1.9586	1.9094	1.8772	1.8423	1.8058	1.7683	1.7300
29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2782	2.2229	2.1768	2.1045	2.0279	1.9446	1.8951	1.8628	1.8277	1.7909	1.7531	1.7144
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646	2.0921	2.0152	1.9317	1.8819	1.8494	1.8141	1.7770	1.7389	1.6999
40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240	2.0772	2.0035	1.9249	1.8389	1.7874	1.7539	1.7172	1.6783	1.6380	1.5963
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401	1.9926	1.9174	1.8364	1.7472	1.6934	1.6584	1.6197	1.5784	1.5352	1.4900
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0867	2.0164	1.9586	1.9105	1.8337	1.7505	1.6577	1.6011	1.5640	1.5228	1.4785	1.4309	1.3797
∞	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307	1.7522	1.6664	1.5705	1.5107	1.4710	1.4260	1.3770	1.3229	1.2622

f – distribution table (at α = 0.1)

V_2	V_1: 1	2	3	4	5	6	7	8	9	10	12	15	20	25	30	40	60	120	∞
1	39.863	49.500	53.593	55.833	57.240	58.204	58.906	59.439	59.858	60.195	60.705	61.220	61.740	62.054	62.265	62.529	62.794	63.061	63.328
2	8.5263	9.0000	9.1618	9.2434	9.2926	9.3255	9.3491	9.3668	9.3805	9.3916	9.4081	9.4247	9.4413	9.4512	9.4577	9.4662	9.4747	9.4833	9.4918
3	5.5383	5.4624	5.3908	5.3426	5.3092	5.2847	5.2662	5.2517	5.2400	5.2304	5.2153	5.1997	5.1835	5.1739	5.1677	5.1606	5.1535	5.1463	5.1391
4	4.5448	4.3246	4.1909	4.1072	4.0506	4.0098	3.9790	3.9549	3.9357	3.9199	3.8955	3.8700	3.8439	3.8284	3.8182	3.8068	3.7953	3.7838	3.7722
5	4.0604	3.7797	3.6195	3.5202	3.4530	3.4045	3.3679	3.3393	3.3163	3.2974	3.2682	3.2381	3.2067	3.1888	3.1768	3.1632	3.1495	3.1357	3.1218
6	3.7759	3.4633	3.2888	3.1808	3.1075	3.0546	3.0145	2.9830	2.9577	2.9370	2.9042	2.8702	2.8348	2.8146	2.8009	2.7855	2.7698	2.7541	2.7382
7	3.5894	3.2574	3.0741	2.9605	2.8833	2.8274	2.7849	2.7512	2.7238	2.7012	2.6655	2.6283	2.5896	2.5676	2.5525	2.5356	2.5185	2.5012	2.4837
8	3.4579	3.1131	2.9238	2.8064	2.7265	2.6683	2.6241	2.5890	2.5602	2.5362	2.4981	2.4585	2.4172	2.3937	2.3777	2.3598	2.3416	2.3232	2.3046
9	3.3603	3.0065	2.8129	2.6927	2.6106	2.5509	2.5053	2.4692	2.4394	2.4145	2.3746	2.3330	2.2896	2.2648	2.2478	2.2289	2.2097	2.1903	2.1707
10	3.2850	2.9245	2.7277	2.6055	2.5216	2.4606	2.4140	2.3772	2.3467	2.3210	2.2796	2.2365	2.1914	2.1655	2.1476	2.1279	2.1077	2.0874	2.0668
11	3.2252	2.8595	2.6602	2.5366	2.4515	2.3895	2.3421	2.3046	2.2735	2.2471	2.2046	2.1601	2.1135	2.0866	2.0680	2.0476	2.0267	2.0056	1.9842
12	3.1765	2.8068	2.6055	2.4807	2.3945	2.3317	2.2836	2.2454	2.2137	2.1867	2.1433	2.0976	2.0498	2.0220	2.0028	1.9818	1.9602	1.9384	1.9163
13	3.1362	2.7632	2.5603	2.4345	2.3474	2.2838	2.2351	2.1962	2.1639	2.1364	2.0921	2.0454	1.9964	1.9678	1.9481	1.9265	1.9044	1.8820	1.8592
14	3.1022	2.7264	2.5222	2.3955	2.3077	2.2435	2.1942	2.1548	2.1220	2.0940	2.0490	2.0014	1.9513	1.9221	1.9020	1.8799	1.8572	1.8343	1.8109
15	3.0732	2.6952	2.4898	2.3621	2.2737	2.2089	2.1592	2.1192	2.0860	2.0575	2.0119	1.9635	1.9124	1.8826	1.8621	1.8395	1.8163	1.7928	1.7689
16	3.0481	2.6682	2.4618	2.3333	2.2443	2.1790	2.1288	2.0883	2.0547	2.0259	1.9797	1.9305	1.8785	1.8482	1.8273	1.8043	1.7807	1.7567	1.7322
17	3.0262	2.6446	2.4374	2.3082	2.2187	2.1530	2.1023	2.0614	2.0274	1.9982	1.9515	1.9016	1.8488	1.8180	1.7968	1.7733	1.7493	1.7249	1.7000
18	3.0069	2.6239	2.4159	2.2860	2.1961	2.1300	2.0789	2.0376	2.0033	1.9738	1.9266	1.8761	1.8225	1.7912	1.7697	1.7459	1.7215	1.6967	1.6713
19	2.9896	2.6053	2.3967	2.2662	2.1759	2.1094	2.0579	2.0162	1.9817	1.9519	1.9042	1.8531	1.7988	1.7671	1.7453	1.7211	1.6964	1.6711	1.6454
20	2.9741	2.5887	2.3797	2.2486	2.1580	2.0911	2.0393	1.9973	1.9625	1.9325	1.8843	1.8326	1.7776	1.7455	1.7234	1.6989	1.6739	1.6483	1.6222
21	2.9601	2.5737	2.3642	2.2327	2.1418	2.0747	2.0226	1.9803	1.9452	1.9150	1.8663	1.8141	1.7584	1.7259	1.7036	1.6788	1.6534	1.6275	1.6010
22	2.9474	2.5602	2.3502	2.2183	2.1271	2.0597	2.0074	1.9648	1.9295	1.8991	1.8500	1.7973	1.7410	1.7082	1.6856	1.6606	1.6349	1.6087	1.5819
23	2.9359	2.5480	2.3375	2.2052	2.1138	2.0461	1.9936	1.9508	1.9153	1.8847	1.8352	1.7820	1.7251	1.6920	1.6691	1.6439	1.6180	1.5914	1.5643
24	2.9254	2.5369	2.3260	2.1933	2.1016	2.0337	1.9809	1.9379	1.9022	1.8715	1.8216	1.7680	1.7105	1.6771	1.6540	1.6286	1.6024	1.5756	1.5482
25	2.9158	2.5267	2.3155	2.1825	2.0906	2.0225	1.9695	1.9263	1.8904	1.8595	1.8092	1.7551	1.6971	1.6634	1.6401	1.6145	1.5881	1.5610	1.5333
26	2.9070	2.5174	2.3059	2.1726	2.0805	2.0122	1.9590	1.9156	1.8795	1.8485	1.7978	1.7433	1.6849	1.6509	1.6274	1.6016	1.5750	1.5477	1.5197
27	2.8990	2.5089	2.2971	2.1636	2.0713	2.0027	1.9494	1.9058	1.8696	1.8384	1.7874	1.7325	1.6737	1.6394	1.6157	1.5897	1.5629	1.5354	1.5072
28	2.8916	2.5011	2.2890	2.1552	2.0627	1.9940	1.9405	1.8967	1.8604	1.8291	1.7778	1.7226	1.6634	1.6288	1.6049	1.5787	1.5517	1.5240	1.4956
29	2.8848	2.4938	2.2814	2.1474	2.0547	1.9858	1.9321	1.8882	1.8518	1.8203	1.7688	1.7133	1.6538	1.6189	1.5949	1.5685	1.5413	1.5134	1.4848
30	2.8786	2.4871	2.2743	2.1401	2.0472	1.9781	1.9243	1.8802	1.8436	1.8120	1.7602	1.7045	1.6447	1.6096	1.5854	1.5589	1.5315	1.5035	1.4746
40	2.8354	2.4404	2.2260	2.0906	1.9967	1.9267	1.8720	1.8270	1.7896	1.7574	1.7042	1.6467	1.5844	1.5478	1.5225	1.4947	1.4659	1.4361	1.4053
60	2.7911	2.3931	2.1770	2.0403	1.9451	1.8740	1.8182	1.7721	1.7337	1.7006	1.6458	1.5862	1.5210	1.4825	1.4560	1.4266	1.3959	1.3639	1.3305
120	2.7478	2.3473	2.1296	1.9916	1.8951	1.8228	1.7658	1.7185	1.6789	1.6447	1.5880	1.5260	1.4574	1.4168	1.3888	1.3574	1.3243	1.2895	1.2528
∞	2.7055	2.3026	2.0833	1.9443	1.8465	1.7730	1.7146	1.6660	1.6250	1.5895	1.5305	1.4655	1.3928	1.3495	1.3195	1.2853	1.2488	1.2094	1.1660